Multi-Hop and Mesh for LoRa Networks: Recent Advancements, Issues, and Recommended Applications

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A comprehensive review is presented on the latest approaches to solutions focusing on multi-hop and mesh LoRa networks through the evaluation of simulations and real-world experiments, based on papers published between 2015 and 2023. The approaches are systematically classified into four (4) aspects: energy awareness, concurrent access and duty cycle regulations, routing protocols, and security. The first aspect encompasses parameter selection, distance, and clustering. The second aspect focuses on network segregation and time synchronisation. The third aspect covers proactive, reactive, and hybrid routing protocols. The aspect of security includes privacy and Denial-of-Service (DoS). Findings show a consensus that multi-hop networks and adherence to radio duty cycle are able to lower average power consumption. Concurrent access method can improve network performance, but more research is needed due to different conclusions. Reactive protocols have a slight advantage over proactive protocols in terms of node deployment scalability and lower power consumption due to less frequent routing table updates. Hybrid protocols are only beneficial in specific topology deployments. Security is still a major concern for LoRa networks, particularly on attacks such as Man-In-The-Middle (MITM), spoofing, wormholes, flooding, packet replay, and information disclosure. The upcoming trend of implementing machine learning algorithms to multi-hop and mesh LoRa networks opens up a wide range of possibilities, ranging from airspace efficiency, efficient route selection, and improving data throughput. A detailed discussion of the aspects' issues and recommended industry applications that will benefit from multi-hop and mesh LoRa networks are also provided.

CCS Concepts: • **General and reference → Surveys and overviews;**

Additional Key Words and Phrases: LoRa, LoRaWAN, multi-hop, energy consumption, duty cycle, routing, security

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1 INTRODUCTION

The recent rapid growth of Internet-connected devices has given rise to countless implementation possibilities and new challenges. These devices form the concept of **Internet-of-Things (IoT)**. IoT is a broader concept that encompasses the network of interconnected physical objects or "things" embedded with sensors, software, and connectivity capabilities. IoT involves various technologies, protocols, and applications that enable these devices to collect and exchange data over the Internet or other networks. The potential of IoT is huge and can be implemented in various fields, such as smart cities [10, 13, 54], smart metering [58], industrial [43], agriculture [46], communications [55], logistics [48], and environmental monitoring [16, 85, 87]. Demand for deploying efficient and cost-effective IoT networks are becoming even more apparent due to the constant growth of globally connected devices and ever-growing variation of application requirements. In 2021, a worldwide total of 11.3 billion IoT devices are connected, and by 2030 there will be an estimated total of 29.4 billion [74]. There are several existing technologies that allow IoT devices to communicate, such as physical Ethernet cabling, powerline network, Wi-Fi, mobile packet services (e.g., **Long-Term Evolution (LTE)/5th Generation Network (5G)**), and satellite. Despite these technologies being mature and stable enough for urban environments, disadvantages still exist and are more pronounced when deployed in remote environments. An example is signal coverage, as most only offer extensive coverage in urban environments. One popular solution is **Low Power Wide Area Networks (LPWANs)** (Figure 1).

LPWAN technology is a subset of IoT that is specifically designed to address the unique requirements of IoT applications. They provide long-range communication capabilities, enabling IoT devices to transmit data over considerable distances, often spanning kilometres, while consuming minimal power and at the expense of throughput [50]. However, it may still be sufficient for wireless remote sensors, **machine-to-machine (M2M)** communications, and future IoT sensors developments. LPWAN protocols generally utilise phase shift keying, frequency shift keying, frequency/time division, or **chirp spread spectrum (CSS)** modulations. LPWAN devices typically connect to the Internet through **gateways (GW)**. These GWs act as intermediaries, receiving data from the IoT devices and relaying it to the Internet. The GWs enable IoT devices to access Internet-based services, transmit data to cloud platforms for storage and analysis, and communicate with other devices or applications within the IoT ecosystem. There are also other types of proprietary modulations such as **random phase multiple access (RPMA)**, developed by Ingenu.

Table 1 is a summarised list of common LPWAN technologies. Taking LoRa, SIGFOX, WEIGHTLESS-P, **Wireless Smart Utility Network (Wi-SUN)**, and Ingenu as examples, these operate on the unlicensed sub-GHz and 2.4-GHz **Industrial, Scientific and Medical (ISM)** spectrum, allowing self-built networks to be built with low deployment time [49]. There are also regional regulations on how radio-emitting devices are supposed to operate on the ISM band, particularly on usable frequency band and duty cycles. Ingenu, however, does not need to comply with any duty restrictions as there are none for 2.4GHz's spectrum. **LTE Machine type communication (LTE-M)** and **Narrowband Internet of Things (NB-IoT)** fall into another category, technologies that utilise the licensed spectrum. LTE-M is a telecommunication standard developed by **the 3rd Generation Partnership Project (3GPP)** for the purpose of IoT M2M

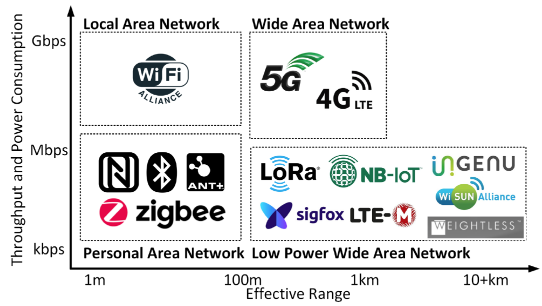

Fig. 1. IoT technologies in relation to range, throughput, and power consumption.

Table 1. List of LPWAN Technologies

Technology	**LoRa** [77]	**SIFGOX** [78]	**LTE-M** [64]	**NB-IoT** [64]	**WEIGHTLESS-N/P/W** [8]	**Ingenu** [39]	**Wi-SUN** [61]
Ownership	Proprietary Physical Layer (PHY) Open Media Access Control (MAC)	Proprietary	Open	Open	Open	Proprietary	Open
Spectrum License	Unlicensed	Unlicensed	Licensed	Licensed	N/P: Unlicensed ISM W: Licensed Television White Space (TVWS)	Unlicensed	Unlicensed
Frequency Band	Sub-1-GHz ISM, 2.4-GHz ISM	Sub-1-GHz ISM	Cell	Cell	ISM or Ultra High Frequency (UHF) TVWS	ISM 2.4 GHz	Sub-1-GHz ISM, 2.4 GHz ISM
Modulation Used	CSS	Binary Phase Shift Keying (BPSK)	BPSK, Quadrature Phase Shift Keying (QPSK), 16 Quadrature Amplitude Modulation (16QAM), 64QAM	BPSK, QPSK	N: BPSK P: Frequency Division Multiple Access (FDMA), Time Division Multiple Access (TDMA) W: Time Division duplexing (TDD)	RPMA	Offset QPSK (O-QPSK) (2.4 GHz), BPSK (Sub 1 GHz)
Frequency	433, 868, 915 MHz, 2.4 GHz	868, 915 MHz	700-2600 MHz	700-2100 MHz	ISM: 868, 915 MHz TVWS: 400-800 MHz	2.4 GHz	868 MHz, 915 MHz, 2.4 GHz
Bandwidth (per channel)	125, 250, 500 kHz	100 Hz (EU), 600 Hz (US)	1.4 MHz	200 kHz	12.5 kHz, 100 kHz	1 MHz	200, 400, 600, 800, 1,200 kHz
Estimated Distance Coverage	Up to 15 km	10 to 40 km	11 km	15 km	Up to 2 km	Up to 48 km	Up to 4 km
Standards	LoRaWAN	Proprietary	3GPP 12	3GPP 13	Weightless Alliance	Proprietary	Wi-SUN Alliance
Security	AES 128	AES 128	LTE Security	LTE Security	AES 128/256	AES 128	AES 128

communications. Compared to other LPWAN technologies, LTE offers a high data rate, voice over network but with drawbacks such as high bandwidth consumption and cost and shorter battery life. LTE-M is a lighter version of LTE with several redundant features removed that are not required for IoT networks while also being able to co-exist within conventional LTE's bandwidth, resulting in lower deployment and operational costs when compared to a full-featured LTE [2].

NB-IoT is also a standard from 3GPP that focuses on indoor coverage. It is a subset of LTE with limited bandwidth and reduced feature sets when compared to conventional LTE, including protocol modifications to cater for low-cost implementations, high sensor density, and long battery life

[59]. Another notable technology is Wi-SUN, a rival to LoRa. This standard enables large-scale outdoor IoT networks including Advanced Metering Infrastructure, Field Area Networks, and Home Area Networks. Wi-SUN's PHY layer is based on the IEEE 802.15.4g standard and IEEE 802.15.4e standard for its MAC layer. Compared to other IoT technologies, LoRa still has the lowest barrier of entry due to its low-cost and easy deployment, thus making it the most suitable technology to implement and deploy a low-power **Wireless Sensor Network (WSN)**. LoRa has an advertised range of up to 15 km in rural, 2 km in urban areas, and low data rates from 0.3 to 50 kbps [72]. As it operates on the globally recognised ISM band, this automatically qualifies LoRa for global use. Besides that, independent private networks can be established without the hassle of commercial licensing and carrier fees and are free to manage their own subset of IoT devices.

1.1 Motivations

Previous researches have uncovered great potential in real-world solutions utilising multi-hop and mesh LoRa. However, proposals and surveys (shown in Table 2) on addressing the challenges of deploying these networks are still somewhat lacking. The technology can offer huge potential for the **Industrial Internet of Things (IIoT)** industries, where connected devices are attached to or powering highly beneficial and productive hardware. Without an active connection, any data being collected and transmitted by the sensors will have no value. LPWAN technology, designed specifically for M2M communications and IoT devices, is defined by its name to enable low power consumption, and long-range wireless connectivity. This is ideal for solutions with limited mobility that lack a suitable power source for recharging. The nature of this technology explains why it is being hailed as a game changer for IIoT [79]. It will drive the growth and adoption of connected systems, as well as improving the overall outlook for the entire industry. More importantly, LPWANs stand to bring just about every business or organisation into the modern age. Haxhibeqiri et al. [34] created a **Strengths, Weaknesses, Opportunities, and Threats (SWOT)** analysis on the existing reliability mechanisms of LoRa/LoRaWAN and identified that more studies are required particularly with security on LoRaWAN protocol backwards compatibility as this will cause patched security flaws to resurface.

Table 2. Recent Surveys Related with LoRa and LoRaWAN

Ref.	Author(s)	Year	Technology	Parameters Evaluated	Contributions	Gaps/Future Directions	Cited Ref.
[34]	Haxhibeqiri et al.	2018	LoRa, LoRaWAN	Path loss, power consumption, throughput, delay	SWOT analysis on security and reliability mechanisms of LoRa and LoRaWAN between 2015 to 2018	More security analysis needed with regards to LoRaWAN protocol backward compatibility causing flaws to resurface	132
[10]	Andrade and Yoo	2019	LoRa	Spreading Factor (SF), Bandwidth (BW), throughput, payload, Packet Delivery Ratio (PDR), Signal-to-Noise Ratio (SNR), range, power consumption	Identify the technological or practical barriers for smart city IoT developments	IoT–LoRa proposals are loosely integrated, not within smart city's development macro-plan	97
[81]	Sundaram et al.	2020	LoRa	Range, PDR, security, power consumption	Link coordination, resource allocation, security, and technical challenges of deploying higher performance LoRa networks	Security needs for specific applications need further study	137
[24]	Cotrim and Kleinschmidt	2020	LoRaWAN	Device type, radio utilisation, energy constraints, type of topologies	Multi-hop proposals for LoRaWAN; comparative analysis and classification, characteristics, and network topologies	No standardisation on LoRaWAN mesh networks	61
[67]	Osorio et al.	2020	LoRa, LoRaWAN	Routing protocols, number of nodes, number of hops, Line-of-Sight (LoS), Non-Line-of-Sight (NLoS), PDR	Approaches to building multi-hop communication in LoRa networks, and routing protocols.	More studies are needed with larger testbeds, lack of security addressing, and techniques to improve power consumption efficiency	15

(Continued)

Table 2. Continued

Ref.	Author(s)	Year	Technology	Parameters Evaluated	Contributions	Gaps/Future Directions	Cited Ref.
[18]	Centelles et al.	2021	LoRa, LoRaWAN	Scenarios, approaches, validation, outcomes, limitations	Implementation of multi-hop and mesh solutions for LoRa and LoRaWAN. Addressed challenges to overcome for achieving a decentralised LoRa network	Co-deployment with shorter range but higher throughput technologies to increase bandwidth. More security research is required.	20
[47]	Lalle et al.	2021	LoRaWAN	PDR, number of nodes, routing methods, number of hops, distance, duty cycle, payload size	Review of routing protocols for LoRaWAN multi-hop networks and comparison between them.	Lack of energy-saving mechanisms, lack of security and privacy approaches in routing, and network scalability.	44
[9]	Almuhaya et al.	2022	NB-IoT, SigFox, Telensa, RPMA, LoRa, LoRaWAN	Power lifetime, network capacity, QoS, cost, security	Simulation tools for analysing LoRa/LoRaWAN network performance, energy consumption, network scalability, coverage, QoS, and security.	Device interoperability issues (co-existence) lacking, possibility for centralised network management	119
[49]	Li and Cao	2022	LoRa	SNR, node distance, Packet Error Rate (PER), PDR, throughput, energy consumption	Research issues on PHY, MAC and application layer. LoRa networking performance metrics including range, throughput, energy, and security.	Network optimisation techniques to further improve LoRa throughput such as compression. Deep AI to promote self-learning and self-healing networks.	134
[80]	Sun et al.	2022	LoRa	Simulator features, testbeds, analytic models, configuration parameters, vulnerabilities, countermeasures	Analysis on LoRa communication studies in terms of PHY and MAC layers, security vulnerabilities and countermeasures, and LoRa-enabled applications	Further improvement of LoRa PHY/MAC stack, more robust network management systems, network aggregation and AI enhancement.	190
[41]	Jouhari et al.	2023	LoRaWAN	Collision avoidance mechanisms, interference management schemes	Analysis on the issues of LoRa signal interference during simultaneous transmissions, gateway densification, and investigation on existing collision avoidance mechanisms	Joint Transmit (TX) scheduling and SF assignment, efficient interference cancellation schemes and carrier sensing, improved gateway densification methods, relay topology with End Node (EN) fairness, and machine learning for MAC	274
Proposed study		2023	LoRa, LoRaWAN	Power consumption, PDR, Received Signal Strength Indicator (RSSI), duty cycle, routing protocols, security, machine learning	Approaches to energy aware LoRa deployments, duty cycle implementations for power efficiency or regulatory compliance, routing protocols, security issues, and machine learning related to multi-hop and mesh LoRa networks.	Optimal parameter selection, payload compression/aggregation for non-linear data, network segregation, time synchronisation, efficient routing protocol, backwards compatibility issues, reinforcement learning	89

Andrade and Yoo [10] explored why LoRa IoT development in smart cities is not advancing fast enough. What the authors suggest is that the existing LoRa proposals are too loosely integrated and are not considered in the macro-plan for the development of a sustainable smart city. Sundaram et al. [81] reviewed on LoRa's security, link coordination, resource allocation, and technical challenges. Strategic placement of GWs, and policies such as dynamic **Acknowledge (ACK)** and retransmission mechanisms, need further research to improve network performance. The authors also suggested further studies on security implications for specific LoRa applications. Cotrim and Kleinschmidt [24] reviewed multi-hop proposals for LoRaWAN, including comparative analysis, classification, functionalities, and network topologies. No standardisation was identified for LoRaWAN multi-hop networks, unlike other technologies such as Bluetooth and Wi-Fi. The authors also recommend conducting further research on multi-hop to promote wider adoption. Osorio et al.'s [67] paper claims to be the first survey paper on multi-hop LoRaWAN routing protocols and has identified that further studies are still needed on security and techniques to improve power efficiency. Co-deployment with shorter range but higher throughput technologies such as ANT and Bluetooth to increase bandwidth has been proposed by Centelles et al. [18]. They reviewed multi-hop solutions for LoRa and LoRaWAN, including proposals to achieve a proper, decentralised network. The authors also suggested that more research is required on security. Lalle et al.'s [47] work is an extension of Osorio et al.'s [67] paper, providing a survey on various routing protocols for LoRaWAN multi-hop networks. The authors also suggested further research on energy saving mechanisms and security. Almuhaya et al. [9] surveyed various LPWAN technologies and simulators for performance analysis, energy consumption, **Quality of Service (QoS)**, and security. Further research was suggested on network management and device interoperability, since ISM band may be cluttered. This paper raised a concern that the co-existence of airspace may pose complications such as interference and congestion.

Li and Cao [49] reviewed LoRa networking performance metrics such as range, throughput, energy consumption, and security. Other factors considered were **Open Systems Interconnection (OSI)** layer-specific issues covering PHY and MAC layers. Figure 2 describes a side-by-side comparison between the 7-Layer OSI model and LoRa's stack. The authors proposed deep **Artificial Intelligence (AI)** as a potential research direction due to its ability to enhance self-learning and self-healing capabilities in networks. This approach is particularly valuable for multi-hop, organic IoT networks. To the best of our knowledge at the time of writing, Sun et al. [80] is one of the latest and most comprehensive review papers on LoRa and LoRaWAN. The review consists of simulators, LoRa parameters, test beds, stack layer improvements, security, and applications. Although their work briefly mentions on multi-hop LoRa, there is no substantial coverage on routing. The suggested future research directions also coincide with what the previous papers stated; improvement of protocols and procedures [10], centralised network management [9, 47, 81], machine learning [41, 49], and network aggregation [33].

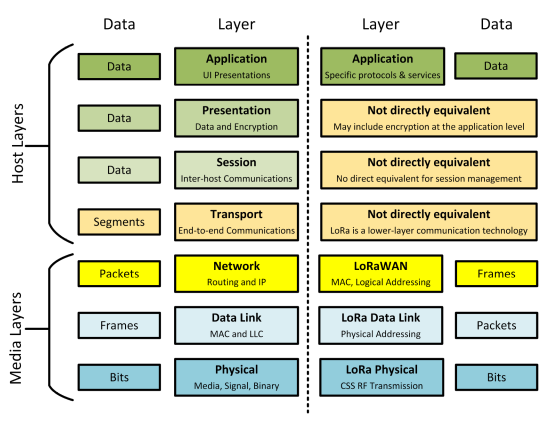

Fig. 2. Comparison between 7-Layer OSI stack and LoRa's [77].

Jouhari et al. [41] surveyed LoRaWAN scalability issues in massive IoT deployments. They analysed works that addresses LoRa signal interference when multiple ENs transmit simultaneously, consisting of SF assignments, interference cancellation mechanisms, and GW densification. The authors also investigated works done on collision avoidance schemes to address concurrent radio transmission issues, consisting of carrier sensing mechanisms, windowing, and multi-hop/relay. However, only clustering works in relation to multi-hop and relay have been analysed. They have highlighted a potential research direction on developing multi-hop routing protocols aiming to provide fairness between ENs in terms of link quality, SF, and energy efficiency to increase the network's total lifetime. They have also suggested jointly designing fair EN routing protocols with GW densification. Recent surveys such as from Li and Cao [49], Sun et al. [80], and Jouhari et al. [41] proposed machine learning as potential research directions. Implementing machine learning techniques in multi-hop LoRa-based networks bring forth a powerful convergence of long-range communication and intelligent data analysis. By integrating machine learning algorithms into these networks, the collected data can be processed and analysed to extract valuable insights and make informed decisions. Machine learning enables the network to potentially detect patterns, predict future trends, optimise routing algorithms, and enhance resource allocation. This integration opens up a wide range of possibilities, from predictive maintenance and anomaly detection to intelligent routing and optimisation in IoT applications. By harnessing the potential of machine learning, multi-hop LoRa networks can greatly enhance their efficiency, scalability, and overall performance, driving advancements in various domains and facilitating the development of smart and connected ecosystems.

While there are previous surveys related to multi-hop LoRa (e.g., Cotrim and Kleinschmidt [24], Osorio et al. [67], Centelles et al. [18], Lalle et al. [47], and Jouhari et al. [41]), those papers only focus on specific approaches and proposals. As to our best of knowledge, we are also unable to find significant survey papers after Lalle et al.'s work that specifically focuses on multi-hop LoRa. As for the other survey papers, multi-hop LoRa has been discussed but only briefly and was not the main topic of discussion. Another difference between this work and others is that a more holistic approach is being presented in terms of the practicality, feasibility, and implementation challenges of multi-hop LoRa networks while incorporating more recent research works. Also, the topics of security and concurrent transmission-related works were not fully presented in the previous papers; these research gaps are pivotal research areas and thus should be addressed accordingly.

1.2 Contributions

A related and interdependent classification is proposed to categorise the aspects that determine the feasibility, functionality, and practicality of multi-hop and mesh LoRa networks: energy awareness, concurrent airspace access and duty cycle, routing protocols, and security. To conserve energy and prolong the lifespan of intermediate sensor nodes, duty cycle regulations can be employed to ensure that nodes are synchronised and alternate themselves between active and idle modes. This helps to reduce energy consumption during idle periods. Energy-awareness can be implemented through the use of energy-efficient parameter combinations, strategic EN placements, clustering, or communication protocols. By combining these two classifications, a network can optimise energy usage and extend node lifetime.

Routing protocols determine the path of data being transmitted between nodes. Selection of a routing protocol can have a significant impact on the security of the network. For instance, proactive routing protocols may be more secure than reactive routing protocols, because they rely on pre-existing routing tables. Energy-efficient routing protocols can also be used to minimise energy consumption during data transmissions. For example, geographic-based routing protocols can forward data packets to the next-hop node based on the node's location, which can minimise the number of nodes that need to be activated during routing. By combining energy-awareness and routing protocols, a network can optimise energy usage while ensuring efficient data transmission. Duty cycle regulations can be employed to ensure that nodes do not consume too much energy while still remaining operational for an extended time. This can help to prevent attacks that drain a node's battery.

Furthermore, security mechanisms such as encryption and access control, can be implemented to prevent unauthorised access to the network, service interruptions, and information disclosure. By combining duty cycle regulations and security, a network can ensure that nodes are operational for an extended period while maintaining a high level of security. Machine learning can complement energy awareness, duty cycle regulations, routing protocols, and security. By analysing historical data and network conditions, machine learning algorithms can identify energy-saving opportunities, adapt routing strategies, detect anomalies, and improve security by identifying patterns of suspicious behaviour.

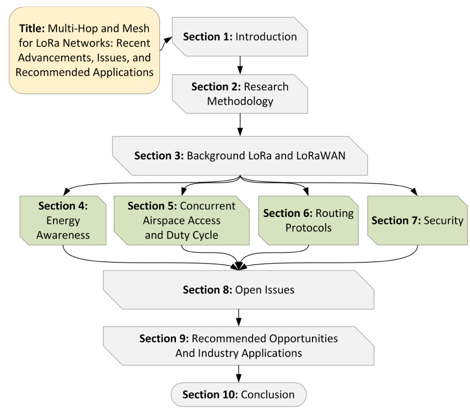

Fig. 3. The structure of this article.

Therefore, this survey article contributes:

(1) A presentation on state-of-the-art solution methodologies and results focusing on specific aspects that determine the feasibility, functionality, and practicality of multi-hop and mesh LoRa networks. This consist of energy awareness, concurrent access and duty cycle regulations, routing protocols, and security.
(2) A classification is proposed to categorise the aspects that determine the feasibility, functionality, and practicality of multi-hop and mesh LoRa networks: (1) energy awareness, includes parameter selection, distance, and clustering; (2) concurrent access and duty cycle regulations, focusing on network segregation, and time synchronisation; (3) routing protocols, covering proactive, reactive, and hybrid routing protocols; and (4) the aspect of security includes privacy, and DoS.
(3) Detailed discussions on the open issues are presented for each of the classified aspects, as well as recommended industry applications that will benefit from multi-hop and mesh LoRa networks.

1.3 Paper Outline

As shown in Figure 3, this article is organised into several sections. Section 2 elaborates the research methodology of this article. Section 3 describes the background of LoRa and LoRaWAN. Section 4 presents the chronological advancements on results that are worth taking notice in relation to energy awareness. Section 5 discusses recent enhancements about radio duty cycle with regards to improved efficiency and adherence to regulations. Section 6 describes current improvements on routing protocols that facilitates a multi-hop and mesh topology. Section 7 outlines current discoveries and solutions on securing LoRa-based networks. Section 8 contains detailed discussions on the open issues for each of the classified aspects. Section 9 provides some potential use cases for multi-hop and mesh LoRa topologies, and, finally, Section 10 provides the concluding remarks.

2 RESEARCH METHODOLOGY

A survey analysis was conducted to shortlist, evaluate, and analyse available and relevant papers. A set of questions were first identified based on the current status and issues related to a working multi-hop or mesh LoRa networks. The questions are as follows:

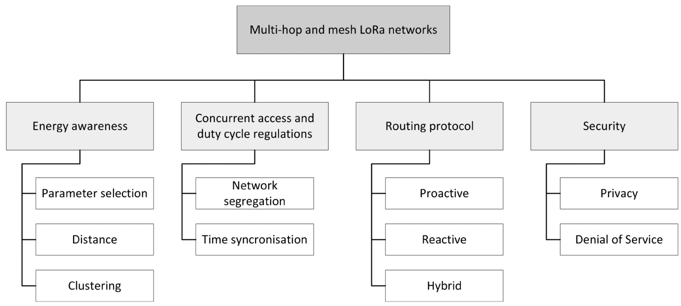

Fig. 4. Classification overview.

(a) What are the initiatives that have been executed to bridge the feasibility gap of multi-hop and mesh LoRa networks?
(b) How to classify the reviewed works into discussable topics?
(c) What applications can be recommended that will benefit from multi-hop and mesh LoRa networks?

To answer these questions, a classification on aspects of interest is determined (refer to Figure 4) and a search was conducted with the following core keywords: *IoT*, *LPWAN*, *LoRa*, *LoRaWAN*, *multi-hop*, *routing*, *duty cycle*, *power efficiency*, *machine learning*, and *security*. The search encompasses all results between 2015 and 2023. The specific year range was chosen as LoRaWAN protocol was only introduced in 2015. ACM Digital Library, IEEE Xplore, ScienceDirect, SpringerLink, and MDPI were chosen as the preferred databases, because both contain the most relevant results. Only English papers were chosen. Duplicates, non-relevant, and those from non-indexed journals were omitted. A total of 89 relevant documents have been identified including technical specification data sheets and certain online resources.

3 BACKGROUND

3.1 LoRa PHY Layer

In 2009, Nicolas Sornin and Olivier Seller intended to develop a long-range, low-power modulation technology for the smart metering industry. Partnering with François Sforza, the trio formed Cycleo. They took CSS, a type of modulation that is mostly used on sonar and adapted it for data transmission. Semtech then acquired Cycleo in 2012 as it was convinced of LoRa's long-range and low power consumption capabilities. LoRa is a PHY layer that uses a modified variant of Frequency-shift keying (FSK), which is patented by Semtech. LoRa utilises a chirp signal with a frequency that changes over time. The data are then encoded onto the chirp signal, resulting in a modulated signal spreading over a wider band beyond the original data signal's bandwidth [4]. LoRa operates within the ISM frequency band, allowing it to be implemented in an open source manner. Each region in the world is designated a usable frequency band (e.g., AS920-923) and number of uplink/downlink channels [63]. The performance of LoRa's CSS modulation is influenced by SF, BW, and carrier frequency. Its CSS modulation scheme implements orthogonal spreading factors, enabling multiple SFs to be simultaneously transmitted on the same channel with minimal degradation on receiver sensitivity. The sensitivity and link budget of a LoRa radio is dependent on the manufacturer's specifications. Some examples include the Semtech SX1272 having a sensitivity down to −137 dBm with a link budget of up to 157 dB [20, 71, 76] and the Semtech SX1276 with sensitivity down to −148 dBm and a link budget of up to 160 dB [75]. The ALOHA channel access mechanism, a collision detection algorithm that transmit messages without pre-empting any collisions is utilised by LoRa. If a sensor wants to transmit a message, then it first decides on a SF, communication channel, and the amount of energy to use. After transmission, the sensor stays open for two reception slots. If it does not receive any message in the first reception window, then it does not open the next slot, reducing power consumption. This type of simplicity leads to low-cost hardware architecture and low power consumption. GWs can simultaneously **Receive (RX)** and TX messages on all channels. LoRa also adopts **Adaptive Data Rate (ADR)**, a power-saving feature by dynamically adjusting data rate and **radio frequency (RF)** output power based on the distance between EN and GW. ADR tries to use the highest possible data rate at all times to reduce the **Time-on-Air (ToA)** of a packet, thus reducing the energy needed to transmit that packet [48].

In November 2020, the LoRa Alliance announced the release of Regional Parameters RP2-1.0.2, which added the support of **Long Range–Frequency Hopping Spread Spectrum (LR-FHSS)** modulation for the EU868, US915, and AU915 regions [6, 86]. Its purpose is to address the challenges arising from the inevitable growth of extensive and densely populated LoRaWAN networks. This is achieved by enhancing network capacity and resilience through the utilisation of **Frequency Hopping (FH)** techniques. The incorporation of intra-packet FH results in improved performance in terms of capacity, sensitivity, and interference mitigation compared to conventional modulation methods that rely on a single channel, all without compromising on the communication range or energy efficiency. Additionally, LR-FHSS is able to tolerate high Doppler rates, making it compatible with Low Earth Orbit satellites. LR-FHSS is intended to only be used for uplink communications, whereas LoRa's CSS modulation remains for downlink communications [65].

3.2 LoRa MAC Layer (LoRaWAN)

LoRaWAN is a MAC protocol utilising Semtech's proprietary LoRa PHY modulation (Figure 5(a)). It defines how devices are supposed to utilise the LoRa hardware, e.g., message formatting, power output, and so on. The LoRa Alliance, established in 2015, is a non-profit organisation tasked to support developments of LoRaWAN and ensuring interoperability standards of all LoRaWAN products. On December 7, 2021, LoRaWAN has been approved as a LPWAN standard by the International Telecommunication Union [7]. The core **network server (NS)** is the root of a LoRaWAN network. Multiple LoRa GWs are then connected to the NS. These GWs then connect to all ENs. Data generated by the ENs are to be received by the GWs and then sent to the NS for data consumption by other end systems. The reverse process occurs for data from the NS intended for the ENs. This architecture leads to a star topology [18]. There are three classes of LoRaWAN (refer to Figure 5(b)), A, B, and C. Class A sensors periodically send packets to the GW and keep its radio open for two short windows, awaiting reply from the GW. It then powers itself down to save energy. The duty cycle also determines the number of packets it can transmit within an hour. Class B sensors are similar to Class A but with pre-programmed receiving time slots. Class C sensors generally have the lowest energy efficiency as the LoRa radio is kept powered on at all times, constantly listening for data packets.

3.3 LoRaWAN Revisions

The LoRaWAN specifications, as defined by the LoRa Alliance, describes any new, modified, or deprecated specifications of LoRaWAN's protocol. The LoRa Alliance also publishes regional specification documents to fulfil various region-specific requirements. Table 3 describes the list of LoRaWAN revisions and highlights. The Version 1.0.x branch continued on in 2018 due to the lower-than-expected adoption rate of version 1.1. Version 1.0.4 will be the last and subsequent revisions will be a continuation of version 1.1's branch.

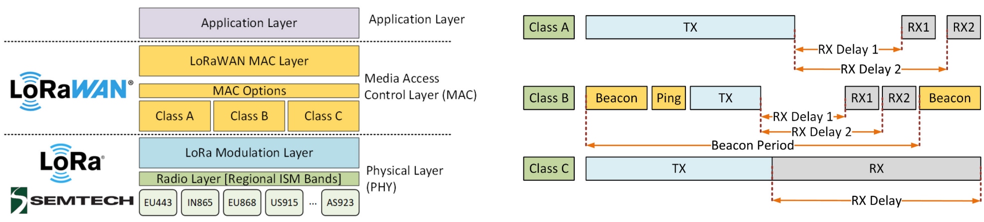

Fig. 5. (a) PHY and MAC layers of LoRa. (b) Different types of LoRaWAN classes and airspace usage.

Table 3. List of LoRaWAN Revisions

Version	Release Date	Highlights	Remarks
1.0	January 2015	Initial Release	
1.0.1	February 2016	Additional descriptions and errata	
1.0.2	July 2016	Segregation of region-specific parameters	
1.1	October 2017	New security features, add roaming capabilities	Deprecated
1.0.3	July 2018	Introduction of Class B	
1.0.4	October 2020	Class B enhancements and security, errata	

4 ENERGY AWARENESS

A low SF increases throughput by reducing the ToA of a packet, which in turn lowers energy consumption. However, lowering the SF can reduce the communication range, because it spreads the signal less, making the radio receivers more susceptible to noise [45]. A high SF allows much longer range, but throughput is significantly reduced due to a longer data packet's ToA [73]. The probability of packets colliding is higher as the data packet stays longer in the air. Battery consumption also increases as the EN will be leaving its radio on for a longer time. Lavric [48] obtained from their simulation results that for an acceptable network performance, PER should not exceed 10%. PER is a ratio between the number of erroneously received packets over the total number of packets. In situations where the EN and intermediary nodes are far away from communication and power infrastructures, efficient utilisation of power is one of the major factors in determining the feasibility of utilising real-world multi-hop and mesh topologies. Table 4 is a list of related works related to the advancements in energy consumption and awareness for multi-hop and mesh LoRa-based networks.

4.1 Parameter Selection

Dix-Matthews et al.'s [26] motivation was to identify the best parameter combination for minimal energy consumption and implementing data compression. Simulation results show that low bandwidths below 125 kHz to increase sensitivity are not worth the extra energy consumption. TX power above 14 dBm significantly increases power consumption because of the activation of an optional chip amplifier. The lowest SF is also preferred as it shortens the packet's ToA. Field tests were performed on a residential area, campus, and farm. A path loss of between 135 and 140 dB was identified to be a good ratio between power consumption and **packet reception ratio (PRR)**. Also with Data-Aware, Resource-Aware lossless compression and **Forward Error Correction (FEC)**, the authors achieved a 16% improvement in **data delivery rate (DDR)** when compared to just increasing SF and TX power. With this knowledge, node deployment strategies for a multi-hop

Table 4. Recent Works Related with LoRa Energy Awareness

Ref.	Author	Year	Classification	Objective	Validation	Metrics Measured	Outcome	Limitation
[26]	Dix-Matthews et al.	2018	Parameter selection	Identify the most efficient combination of parameters to achieve minimal energy consumption	Simulation and real-world	Energy per packet, RSSI, path loss, BW, TX power, SF, DDR	DDR improved by 16 to 25%. FEC with low energy settings is better than increasing packet reception ratio.	Frequency and Code Rate (CR) variations were not considered in the evaluation
[11]	Anedda et al.	2018	Distance	To develop an energy efficient multi-hop solution	Simulation	Power consumption, number of packets, battery capacity, time of operation	Multi-hop provides energy savings of up to 15%	Mesh topology is not considered, only linear multi-hop
[14]	Barrachina-Munoz et al.	2019	Parameter selection	To enable efficient, reliable and low power consumption for far-reached ENs using Machine Learning	Real world	Cycle bottleneck in iteration t, Cumulated bottleneck energy until t	Epsilon Multi-Hop begins to outperform single hop as the learning iterations exceed 110, starting with 7% energy savings.	Parent nodes consume more energy due to relaying child data. Every EN is assumed to be able to communicate with the GW in a single hop manner.
[13]	Aslam et al.	2020	Distance	Evaluate the energy consumption of relay and star-of-stars topology	Real world	PRR, distance, transmit power (mW), time, SF	Multi-hop networks identified to consume less power than single hop.	The placement of intermediate nodes are important to achieve optimal energy efficiency.
[69]	Park et al.	2020	Parameter selection	Evolution of ADR using deep reinforcement learning	Simulation and testbed	SF, TX Power, SNR, Transmission count, arrival rate	Achieved 15% improvement over ADR in terms of the ratio of throughput over energy transmission	Low learning efficiency. GW needs to be computationally more powerful than ENs.
[70]	Paul	2020	Clustering	Evaluate Distance-Ring Exponential Stations Generator (DRESG) framework for a multi-hop network and performance over single-hop networks	Simulation	Energy consumption (J), number of rings, routing process used	Variable Hop is able to balance out power consumption. Packet aggregation extended network lifetime by 1.67 times for a 4-ring setup.	Packet aggregation may induce delays in the time it takes for transmitted data to reach the GW
[32]	Gupta and Singh	2021	Clustering	Evaluate the benefits of clustering by implementing path-based and data-centric multi-hop algorithms	Simulation	Number of Nodes, Sensing Range of Nodes, Total Energy Dissipated	Avg. energy consumption between 3-5 times lower than either direct path to GW or cluster-based multi-hop	Duty cycle regulations not adhered to, packets sent every 2s
[52]	Lumet et al.	2021	Distance	Evaluate the data extraction rate and energy consumption of a two-hop non-line of sight LoRaWAN network	Testbed	SF, energy consumption (J), distance, Data Extraction Rate (DER)	DER improved in urban environments. Energy consumption reduced when compared to a single-hop SF12 link.	Only tested with a two-hop network
[31]	Grochla et al.	2022	Distance	To enable node relaying while keeping in check with the node's battery constraints	Simulation	RSSI, geo distance (m), SF, power consumption (mA), BW, TX power	Between 0.5% to 4% decrease in transmission time, leading to an energy efficiency increase of up to 4%.	Requires centralised network management. Routing is suggested to be a more efficient option for multi-hop communication.
[33]	Gupta and Singh	2022	Clustering	To evaluate an energy-efficient data packet aggregation scheme for clustered LoRa communication	Simulation	Number of Nodes, Energy Consumption(mJ), Aggregation Ratio, Number of Packets Transmitted	Path-based and data-centric aggregation algorithms outperforms traditional LoRa single hop in terms of power consumption	Assumed max payload fixed at 50 bytes and size variations within payload were not considered
[68]	Pandey et al.	2022	Parameter selection	Enhance data throughput, minimise energy consumption for transmitting ENs and reduced TX delays using Q-learning	Simulation and testbed	Network lifetime performance, residual energy, data throughput, data latency, bandwidth utilisation, data interference	After 15000 iterations,only 60% nodes were dead. When compared against 3 other protocols, proposed method ranks best among all metrics.	Q-learning not evaluated on mobile LPWAN. Simulations assume that all ENs and GW are visible to each other.

LoRa network can be improved to ensure optimal placements of ENs that have reliable connection quality.

Barrachina-Munoz et al. [14] pointed out that the computation of energy-efficient optimal-hop routes using trial-and-error and brute-force methods is quite challenging with respect to time complexity and energy constraints over the ENs. To address these issues, the authors investigated the epsilon-greedy method and proposed **Epsilon Multi-Hop (EMH)** to compute the optimal routing path through **Reinforcement Learning (RL)**. They demonstrated that the optimal setting of the epsilon-greedy algorithm resulted in improved energy efficiency. However, the optimal settings of the epsilon-greedy algorithm heavily depends on how the network is designed and deployed. The authors also investigated on how the algorithm affects energy consumption, and validating the hypothesis that multi-hop routing yields significantly reduced energy consumption for LPWANs. EMH begins to outperform single hop as the learning iterations exceed 110, starting at 7% energy savings.

LoRa's use of ALOHA's mechanism has a tradeoff for power consumption and equipment simplicity, higher chances of packet collisions due to longer ToA, and the near-far issue. The process of determining the best parameter combination to achieve minimal energy consumption is one of the factors to maximise power efficiency. With this in mind, LoRa's specification provides a feature known as ADR. ADR's function is to find the best LoRa parameter values to attain optimal efficiency. Park et al. [69] studied the feasibility of applying deep reinforcement learning to further enhance ADR's features. Modelled after a Markov Decision Process, the authors proposed Q-Learning using Deep-Q Network Algorithm and built a simulation model and a mini testbed to evaluate their findings. The proposed method has an average of 15% improvement over ADR in terms of throughput over power consumption. ADR can be beneficial to certain multi-hop LoRa networks such as cluster-based topologies, since the ENs will be able to conserve their energy by automatically adjusting the SF and TX power. The drawback of this method is that GWs or NS are required to have agents to do EN evaluations running in them that can increase the architecture's complexity. The algorithm was also not very efficient, taking up to 50 hours to calculate in a simulation.

Pandey et al. [68] focused on addressing the challenges of energy efficiency and QoS in LPWANs for IoT applications. The paper introduces a novel approach that utilises Q-learning, a RL technique to optimise data routing in LPWANs. The proposed method updates the network's Q-matrix at specific intervals, using it to select a relay device that maximises its cumulative reward value for the route path selected between the EN-GW pairs. Each EN is evaluated based on residual energy, data queue size, channel conditions, and TX power levels. The authors' goal was to minimise energy consumption for ENs during data transmission, reduce transmission delay, and enhance data throughput via efficient bandwidth utilisation. A simulated testbed was conducted to compare the proposed method with other protocols such as Modified **Low Energy Adaptive Clustering Hierarchy (LEACH)**, traditional **Multi-Hop Routing (MHR)**, Learning-based MHR, and single-hop. In terms of network lifetime, after 15,000 iterations (1 iteration refers to a single transmission from an EN) the proposed method only has a total of 60% ENs that have depleted their energy, as opposed to nearly 100% for the other protocols. However, the authors did not evaluate their proposed Q-learning method on mobile ENs.

4.2 Distance

Anedda et al. [11] describe a reliable variant of the traditional star topology whereby an EN is able to transmit to another EN that is closer to the GW; facilitating a linear multi-hop design. The authors call it, an "energy-efficient multi-hop communication solution" (e2McH). It determines the energy consumption and residual battery life of a node and transmits to another EN closer to the GW determined via a distance formula. With this information, e2McH then determines the best next-hop route. A simulation was conducted and results show that when compared to single hop, a multi-hop environment is able to achieve energy savings of around 8–15%. Similar results also have been shown by both Lumet et al.'s [52] and Grochla et al.'s [31] works. Instead of a single long-distance, high SF link, Lumet et al. placed an intermediate **relay node (RN)** and utilised SF7. DER for a two-hop SF7 network improved by up to 8 times when compared to a single-hop SF8 network. As for Grochla et al.'s work, an algorithm identifies potential relays based on the node's geographical location, shuts off any unsuitable relays, and then begins assignments of RNs to ENs that have weak RSSI values. Simulated results show improved energy efficiency through 0.2 to 4% lower transmission times. However, a central controller is needed to orchestrate the algorithm, further complicating the architecture for a multi-hop network. Hence, the authors suggested that routing protocols are still a more efficient way for multi-hop networks. In a nested topology, nodes are separated into several clusters. Each cluster has its own GW. Nodes from the respective clusters only communicate with the designated GW and multi-hop capabilities are on the GWs. This is beneficial, as it can extend node battery lifetime through maximising energy efficiency, as described in a study conducted by Aslam et al. [13]. A field test study shows that the placement of **GW Relay Nodes (GW RN)** is also important to achieve optimal energy efficiency. A GW RN was then placed in between EN and GW, a variable for analysis. To achieve a QoS of 90%, with the GW RN being placed 50 m from the GW, a total of 22 mW is required to transmit the data from EN to GW. Only 16 mW is required when the GW RN is placed 100 m from GW. There is a distance limit, however, as the power consumption went back up to 18 mW when the GW RN is moved to 120 m away from the GW. When compared to single-node operation, nested nodes achieved energy savings of up to 51% for 5 nodes in a nested cluster.

4.3 Clustering

Paul's work [70] proposed a DSREG based multi-hop ring routing scheme forming segregated clusters at various specific radii from the central GW. Three different routing strategies were evaluated; **direct-hop to the GW (DH)**, **next-ring hop (NRH)**, and **variable-hop (VH)** (a combination of DH and NRH). In NRH, nodes in ring 1 consumed the most power, because it serves as the last stop relay for all other nodes in different rings. Simulation results led to an energy efficiency increase of at least 67% in comparison to a single hop. All ENs within a LoRa network communicate to all GWs within the nodes' reach. Gupta and Snigdh's work [32] mentioned that redundant transmissions exist that could lessen the nodes' lifetime due to higher energy consumption. The authors have proposed two types of clustering algorithms and topologies, with reference to the similar success of Zigbee's improvements in network performance. A k-Medoid path-based strategy was proposed on determining the shortest-path multi-hop route to a GW that in turn connects to a cluster head device, connected among a cluster of devices. This ensures all devices within the same cluster are connected to the same shortest distanced GW. The second proposed algorithm utilises a data-centric-based approach; where the shortest path is among nodes that share similarities in the collected sensor data. All nodes that share these data collection similarities are segregated into clusters. Simulation results show that the two proposed algorithms outperformed both single-hop and cluster-based layer approach (similar to Farooq's [29] and Paul's [70] model) in terms of energy consumption. The authors have also suggested that data-centric clusters will have an improved performance when data aggregation methods are used, also to what Paul's work has shown. Gupta and Snigdh [33] continued on their existing work on clustering and evaluated payload aggregation with clustering as a solution to eliminate redundancy, enhance network efficiency, and increase network lifetime. Like compression, payload aggregation is also capable of improving energy efficiency through the reduction of the ToA target. The aggregation ratio and buffer size of a node were used to determine the suitable number of nodes to be aggregated. Link quality factors such as SF, BW, RSSI, and CR were used to generate route costs. With the obtained route costs, an algorithm was proposed to load balance the ENs, preventing them from overloading. Two data collection algorithms have also been proposed: path-based and data-centric. A comparison was

Table 5. List of Regional Frequency ISM Bands and Duty Cycle

Plan Name	Frequency Range	Duty Cycle	Applicable Region
EU863-870	867.1 MHz–869.525 MHz	863.0–868.0 MHz: 1% 868.0–868.6 MHz: 1% 868.7–869.2 MHz: 0.1% 869.4–869.65 MHz: 10%	Europe, Africa
US902-928	903.9 MHz–927.5 MHz	1%	USA, Canada, South America
CN470-510	486.3 MHz–508.1 MHz	1%	China
AU915-928	916.8 MHz–927.5 MHz	1%	Australia
AS920-923 (AS1)	921.8 MHz–923.2 MHz	1%	Japan, Malaysia, Singapore
AS923-925 (AS2)	923.2 MHz–924.8 MHz	1%	Brunei, Cambodia, Hong Kong, Laos, Taiwan, Thailand, Vietnam
KS920-923	921.9 MHz–923.3 MHz	1%	Korea
IN865-867	865.0625 MHz–866.550 MHz	1%	India

done between the two proposed algorithms and traditional LoRa single-hop. Simulation shows that path-based and data-centric outperforms traditional LoRa approach in terms of power consumption when density of nodes increases. However, the simulation only assumed a maximum payload of 50 bytes. Data size variations within the payload were also not considered during the simulation.

5 CONCURRENT AIRSPACE ACCESS AND DUTY CYCLE REGULATIONS

Different regions in the world are governed by their respective duty cycle regulations that LoRa needs to adhere to. The duty cycle in LoRa's context is a ratio of the time a device's radio is on and transmitting over a device's radio being idle. Taking Malaysia as an example, the assigned frequency plan is AS920-923 between 921.8 and 923.4 MHz [63]. The technical specification for **Short Range Devices (SRD)**, and Class Assignment No.1 of 2022, published by the Malaysian Communications and Multimedia Commission, states that devices operating under that frequency range are classified as SRD. SRDs are fixed or mobile stations that have a radio frequency output, with either an external or integrated antenna [23]. ISM devices fall under SRD, which are required to have a maximum TX power of 500 mW **Effective Isotropic Radiated Power (EIRP)** with either a maximum 1% duty cycle, frequency hopping, or **Listen Before Talk (LBT)** [22]. Table 5 is a subset list of regional parameters for ISM devices. Other regions such as Europe are even stricter whereby only a maximum duty cycle of 1% is allowed. A 1% duty cycle translates to a maximum transmission ToA of 36 s per hour (or 864 s per day) per device regardless of whether the transmissions were successful.

Other than adhering to duty cycle just for compliance, it can also be used to cap the power requirements of ENs. This leads to gains in battery lifetime and it has attracted attention from the academic community. One factor that wastes energy is the "idle listening problem," whereby

Table 6. Previous Works Related with Concurrent Airspace and Duty Cycle Regulations

Ref.	Author	Year	Classification	Objective	Validation	Metrics Measured	Outcome	Limitation
[89]	Zhu et al.	2019	Network segregation	Segregate data traffic into several subnets via multiple-access dimension using different SF for enhanced density.	Simulation and real world	PRR, Number of nodes, SF, hops, ToA	ToA of data packets reduced roughly 20% compared to SF7 only network. Mean PRR of each subnetwork is approximately 90% when the TSCA is used.	The algorithm will cause some first nodes to be far away from the sink, higher energy consumption. Not very effective if the area is rectangular shaped.
[1]	Abrardo and Possebon	2019	Time synchronisation	To address multi-hop with time synchronisation in an underground environment	Real world	Distance, time offset, number of nodes, battery depletion rate in mAh	Able to achieve node sleep time of 1 hour with minimal time drift and power savings of up to 50% when compared to non-optimised time sync method.	Despite power savings, this protocol introduces additional data lag.
[28]	Ebi et al.	2019	Time synchronisation	Synchronised LoRa linear mesh via beacon flooding (TDMA).	Simulation and real world	Sync offset, PER, energy consumption millijoules, SF, SNR, RSSI	Mean synchronisation offset obtained was 253 µs. Packet loss for synchronised mesh was at 2.2%.	Max number of child ENs limited to 5 due to LoRaWAN's payload standard (51 bytes). RNs consume more power than ENs causing imbalanced energy depletion.
[15]	Bezunartea et al.	2019	Time synchronisation	To implement Time Slotted Channel Hopping (TSCH) MAC protocol for 1% duty cycle compliance, initially for IEEE 802.15.4-based networks, on LoRa PHY layer.	Real world	SF, BW, offset timings (start of frame, guard time, ACK delays)	Algorithm implemented in Contiki-NG. SF used as basis to calculate the time slot offsets. At SF7 and BW of 125 kHz, the timeslot length calculated is 400 ms.	With a 22 ms guard time, a beacon is needed to resynchronise about every 1000 slots. At 1% duty cycle, TX is only allowed 1/100 slots, meaning 10% bandwidth used just for synchronisation. Only one SF can be selected for the whole network.
[40]	Jiang et al.	2021	Network segregation	Hybrid multi-hop WSN using LoRa and ANT. Utilise a modified TDMA protocol for radio synchronisation and centralised management.	Real world	SF, PDR, PER, Packet Miss Rate (PMR), distance, ToA	SF12 reference nodes achieved 51.5% PDR. All synchronised nodes achieved at least 96% PDR. PMR less than 2%.	TDMA schedule table dissemination increases overhead during setup phase. Max payload size limits the number of nodes.
[37]	Hong et al.	2023	Network segregation	To automatically assign SFs evenly among ENs using RL for the purpose of reducing collisions.	Simulation and real world	Packet Collision Rate	In simulation, compared to LR-opt-pro, LR-greedy, ADR, and EXPLoRa-AT, LR-RL achieved lowest PCR. For real testbed, LoRaMesh used to evaluate LR-RL, LR-random, and ADR with LR-RL achieving the lowest PCR.	Values of 'Learning Rate' and 'Discount Factor' were assumed and not sufficiently explored to achieve the best performance. Long adaptation time causes packet collisions due to initial random transmissions.

an EN stays active but no data are being transmitted or received. Duty cycling reduces energy consumption by reducing this idle listening window. This section discusses related works toward multi-hop/mesh LoRa-based networks on duty cycle and concurrent airspace implementations, as shown in Table 6.

5.1 Network Segregation

Zhu et al. [89] explored the concept of segregating ENs into several subnets, with a sink node on each subnet allocated to a specific SF. This forms several multi-SF clusters that can all concurrently transmit. Initially, all nodes are connected at SF7 and then a self-developed algorithm called **Tree-based SF Clustering (TSCA)** is used to iterate the nodes and split them into different SF groups. Comparing to a standard single SF7 multi-hop network, TSCA managed to reduce the average network ToA by around 20% while maintaining an average PRR of around 90%. However, results show that the algorithm does not work optimally for nodes that are scattered in a rectangle-based area and is considered a possible future research opportunity. Jiang et al. [40] explored multi-hop LoRa and technology co-existence. A long-range multi-hop mesh network using LoRa is first deployed, and then for each LoRa cluster, the mesh network's capability is extended using ANT. ANT is used to build a short-range dense star network for each individual LoRa subnetwork with a coverage radius of 30 m (Figure 6). The TDMA algorithm from Liao et al. [51] was used to propagate time synchronisation broadcasts from the parent LoRa node and the authors also implemented a customised version of Offset-CT. At the parent node, the authors used Ebi et al.'s [28] methodology of acquiring the **Coordinated Universal Time (UTC)** via GPS. Several tests were conducted and results show that the average number of hops is four with SF7, and average PDR between 96 to 98%. A SF12 control node using single hop was also evaluated and it achieved an average PDR or 51.5%. This research indicates that multi-hop networks can be structured without the necessity of adopting a monolithic architecture.

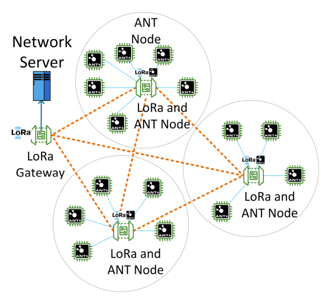

Fig. 6. Technology co-existence: LoRa for wide mesh networks and ANT for small star networks.

Hong et al. [37] proposed a RL-based approach for optimising the allocation of SFs in LoRa networks. The authors first introduce the concept of SF, which determines the data rate and coverage range in a LoRa network and highlight the challenges of SF allocation in terms of network capacity, interference, and power consumption. They then present a RL framework that utilises a Q-learning algorithm to learn an optimal SF allocation policy. The RL agent interacts with the network environment, observing the network state and making decisions on SF assignment to maximise the overall network performance. Three approaches were proposed: LR-opt-pro, where LoRa ENs choose a SF according to a group of probability values; LR-greedy, where ENs select an SF with the objective on maintaining **Packet Collision Rate (PCR)** equilibrium among all SFs; and LR-RL, where ENs select a SF that guarantees reward equalisation based on RL. PCR was used as the performance metric. LR-greedy achieved lower PCR over LR-opt-pro, because LR-greedy contains historical PCR information for each EN transmission. However, the authors stated that LR-greedy's strategy at load balancing SFs is coarse-grained, which led the authors to further develop LR-greedy into LR-RL. Figure 7 is a general process flow of the LR-RL algorithm.

The low PCR of LR-RL also outperforms ADR and EXPLoRa-AT in single-hop simulations. As for real testbed, LR-RL and LR-Random were evaluated in a LoRa-Mesh network with LR-RL achieving the lowest PCR, just under 6% for LR-RL when compared to an average of 18% for LR-Random. ADR was not evaluated in LoRa-Mesh as it has no similar SF allocation methods. Furthermore, the parameters of "Learning Rate" and "Discount Factor" for the proposed algorithms were given fixed values and the authors have admitted that they have not sufficiently explored the various possible combinations to achieve the best performance. Adaptation time for new LoRa nodes can also cause packet collisions due to random transmissions.

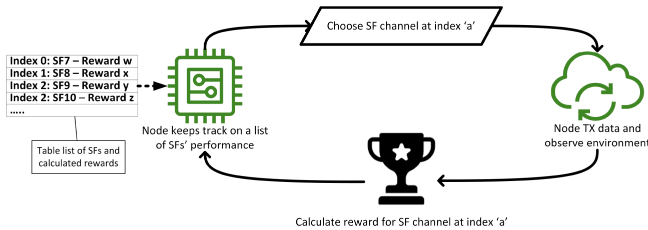

Fig. 7. General process flow of the LR-RL algorithm.

5.2 Time Synchronisation

Ebi et al. [28] worked on time synchronisation and duty cycle for underground deployments. They proposed a synchronised LoRa mesh architecture using beacon flooding and extending its capabilities using **repeater nodes (RpN)**. The TDMA protocol is modified to synchronise and allocate RX slots instead of the usual TX ones. This approach reduces energy consumption as ENs only need to wake up once in receive mode during their respective timeslot. A parent RpN is selected to administer a fixed communication cycle that is subdivided into time slots. First, a beacon packet is transmitted containing the current system time. Upon beacon reception at the next EN, it calculates the latencies of processing time, ToA, and forwarding window to retrieve the original time. The average time synchronisation offset obtained was 253 μs. Synchronisation offset is defined by the time difference between a RpN and EN. The packet loss rate recorded was around 2.2%. A **Linear Sensor Network (LSN)** is a scenario whereby ENs are arranged in a daisy chain. Each node either transmits or forwards the data to an adjacent node. Abrardo and Pozzebon [1] have built an underground testbed in the Aqueducts of Sienna, Italy, to evaluate the energy efficiency using duty-cycling and encountered the challenge of synchronising all ENs to achieve accurate cooperative transmissions. They proposed segregating the time of a node's operation to Synch, Data, and Sleep periods and then developed a protocol to synchronise the ENs clocks in the Synch period. This enables the ENs to accurately only wake up during transmit or receive timeframes. The testbed environment has an average length of 15 km, with nodes spaced apart by around 200 m with nodes on SF12, 125-kHz BW for maximum range. Without any constant time synchronisation, the maximum sleep time achieved without packet loss is 120 s. With the proposed protocol, a sleep time of 1 hour was achieved with a maximum achievable power reduction of 50%. Both experimental results from Ebi et al. and Abrardo and Pozzebon show that time synchronisation among ENs for the purpose of node energy efficiency and adherence to duty cycle regulations during data relaying are possible without the assistance of external technologies such as DCF77 or GPS for all ENs.

Bezunartea et al. [15] evaluated TSCH and duty cycling compliance on multi-hop LoRa, a MAC protocol originally specified for IEEE 802.15.4 networks. The author's goal is to address the challenges of channel coordination and synchronising the power on timing of the LoRa EN radios. Airtime for each EN is split into timeslots (illustrated in Figure 8). To adapt TSCH for multi-hop LoRa, these timings first need to be adapted, which is one of the authors' goals. The authors we are able to tackle most of the challenges that multi-hop LoRa created; a time division mechanism to synchronise wake up times for the nodes, efficient use of airpsace by diversifying transmissions over a range of frequencies that keeping in mind of the regional ISM regulations. However, as

LoRa's throughput is much lower than IEEE 802.15.4, the authors deduced the timeslots for LoRa will have to be significantly larger. At SF7 and 125-kHz BW, the determined minimum timeslot length required for each LoRa EN is 400 ms; and with a 22 ms guard time, a beacon is needed to resynchronise about every 1000 slots. Adhering to the 1% duty cycle, TX is only allowed 1/100 slots, meaning 10% bandwidth used just for synchronisation. Being that the current implementation only allows one SF to be selected for the whole network, the authors intend to conduct further investigations to leverage multiple SFs within the same network.

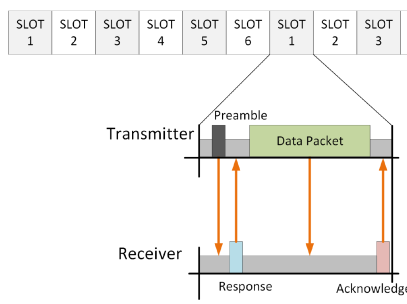

Fig. 8. TSCH: For each timeslot, a determined pair of transmitter and receiver goes online for data transfer.

6 ROUTING PROTOCOLS FACILITATING MULTI-HOP AND MESH TOPOLOGIES

Some routing protocols can potentially be applied, ported, and innovated from other standards such as Bluetooth and Zigbee; enabling LoRa-based networks to go beyond its original architectural design. Studies have suggested that multi-hop reduces power consumption and improves coverage [11, 13, 31, 32, 52, 70]. As Internet infrastructure coverage is one of the challenges, routing protocols will definitely contribute to the lowering of entrance barrier for mainstream adoption as the deployment cost of GWs will decrease. LoRa long-distance links do experience packet losses from obstacles or radio interference e.g., buildings, hills, and trees. These obstacles create NLoS conditions that significantly increase packet loss. Radio interferences such as non-LoRa devices operating on the same ISM channel may degrade signal quality and thus reduce effective range. Multi-hop routing schemes emerge as alternative solutions to improve EN connectivity. These solutions also increase the potential of new applications, making LoRa an active competitor in the IoT arena. There are three major categories of routing protocols: proactive, reactive, and hybrid [53]. Table 7 lists the recent works done on routing protocols for LoRa networks.

Table 7. Previous Works Related with Multi-hop Routing Protocols in LoRa-based Networks

Ref.	Author	Year	Proposed Name	Classification	Topology	Objective	Validation	Metrics Measured	Outcome	Limitation
[38]	Huang et al.	2018	Modified Ad-hoc On-demand Distance Vector (AODV)	Reactive	Linear	Test AODV on LoRa	Simulation	PDR, RSSI	Multi-hop up to 13.6 km with 99% PDR. Single hop reliable up to around 1.6 km.	LoRa constraints not considered during route formation and RX/TX.
[53]	Lundell et al.	2018	Hybrid Wireless Mesh Protocol (HWMP) and AODV	Hybrid	Linear	Transmission efficiency and node lifetime.	Simulation and real-world	RSSI, Route construction duration	Able to uplink messages traversing a four-hop network.	Route failure handling, errors, timers, and ACKs have yet to be implemented.

(Continued)

Table 7. Continued

Ref.	Author	Year	Proposed Name	Classification	Topology	Objective	Validation	Metrics Measured	Outcome	Limitation
[42]	Kawabata et al.	2019	Convergecast	Hybrid	Tree	Assess Convergecast and Adaptation Duty Cycle Control (ADCC) using a combination of synchronised and asynchronised protocols	Simulation and real-world	RX/TX energy, node count, node lifetime (h), to-sink delay (s)	Near nodes use synchronous, far nodes use asynchronous. Compared to ADCC, TX delay and lifetime increased by around 1s and 6 times, respectively.	Synchronised protocol saves more power but required accurate time sync and far nodes have a higher chance of sync errors.
[27]	Dwijaksara et al.	2019	Gateway-to-Gateway (G2G)	Reactive	Tree	Enable G2G hopping	Simulation	ToA, throughput, round-trip delay, downlink packet timeout	G2G's ToA route metric improved throughput, faster links selected. LBT reduced packet collisions.	Buffering, downlink timeout ratio and round-trip delay increased due to potential congestion at intermediate GWs.
[17]	Carvajal-Gomez and Riviere	2020	RoVy	Reactive and Active	Various	Multiple routing protocols on same heterogeneous LPWAN network	Simulation	Round-trip latency, route discovery delay	Round-trip packet latency around 110 ms, 95% improvement.	Interoperable nodes acting as GW are potential points of failure.
[25]	de Farias Medeiros et al.	2020	Distance Vector Routing (DVR), AODV, and Dynamic Source Routing (DSR)	Reactive and Active	Tree	Comparison of DVR, AODV and DSR	Simulation	PDR, round-trip delay, throughput, power consumption	Reactive better for dynamic networks with high throughput, proactive protocol for static.	Said protocols unable to cope with high speed mobile nodes due to infrequent route updates.
[88]	Yang et al.	2021	FerryLink	Reactive	Linear	Enable on-demand packet relaying	Real-world	Loss rate, detection accuracy	The loss rate is lower than 1.6%, while the detection accuracy is above 98.8%.	The experiment only evaluated two hops. Unknown for more than two hops.
[36]	Hong et al.	2022	Hierarchical-based Energy Efficient (HBEE)	Reactive	Mesh	Evaluate PER and node lifetime in comparison to AODV	Simulation and real-world	PDR, PRR	Better power efficiency, lower round-trip delay. 95.7% and 89.8% PRR on four-hop test during network formation and normal operation.	Current implementation has all radios constantly powered on.
[66]	Osorio et al.	2022	MACGSP6	Reactive	Mesh	Apply gossip-based routing and evaluate end-to-end delay	Real-world	PDR, Hop Count, Round-trip Delay	PDR improved from 0% (single hop) to a 53% in the same network with NLoS conditions.	Every hop adds 3 s delay. Not recommended for time sensitive data.
[30]	Feng et al.	2022	MLoRa	Reactive	Tree	Compare single hop with Weight Superposable Minimum Spanning Tree (WSMST) algorithm on node lifetime and packet ToA	Simulation	Gain over cost ratio	Node lifetime increased 10-50%. Node lifetime's Standard deviation more uniform due to balanced power usage.	Number of routes limited due to payload size. Backtracking process induces processing delay.
[3]	Ahmar et al.	2022	Smart-Hop	Hybrid	Tree	Compare single hop with low-overhead routing protocol while maintaining range	Real-world	SF, Extraction Time	Data extraction time decreased between 8% (worst) to 64% (best)	Data aggregation only beneficial at reducing delay if data are of a similar type.

(Continued)

Table 7. Continued

Ref.	Author	Year	Proposed Name	Classification	Topology	Objective	Validation	Metrics Measured	Outcome	Limitation
[62]	Muthanna et al.	2022	Multi-hop Routing for QoS-aware LoRa (MQ-LoRa)	Reactive	Tree	Evaluate QoS-aware protocol for node lifetime, and performance gains in comparison to RT-LoRa, multi-hop relay and LoRa+	Simulation	PRR, Energy Consumption, Round-trip Delay, Throughput	MQ-LoRa achieved up to 28% increase in energy efficiency, and PRR increase of up to 40%.	Mobile ENs not considered in evaluation. Performance gains only maximised when multiple GWs are available.

6.1 Proactive Routing Protocols

de Farias Medeiros et al. [25] evaluated DVR, AODV, and DSR, the first a proactive protocol, and the latter two are reactive. Cupcarbon was used to simulate 25 ENs in a city topology. The parameters evaluated were PDR, average end-to-end data delay, throughput, and power consumption. For DSR, a predefined duration of 5 minutes was set to trigger the sharing of routing tables. For AODV, the route maintenance mechanism was set to repeat every 30 s. Simulation results show that reactive protocols provide better performance when the network topology is dynamic and fast-moving, with AODV having the highest PDR, followed by DSR. At stationary or low speeds (under 20 km/h), DVR has the lowest end-to-end delay. However, DVR encountered high PER at high moving speeds, since the routing table is only refreshed every 5 minutes, invalidating the routing table entries as the ENs move out of RX/TX range. AODV has the highest power consumption but that was due to the frequent route maintenance process. Carvajal-Gómez and Rivière [17] addressed the issues of heterogeneous **Mobile Ad hoc Networks (MANETs)** through adaptive routing using reactive AODV and proactive **Destination Sequenced Distance Vector (DSDV)**. Called RoVy, it allows ENs to automatically switch between these two protocols depending on the environment. Dense zones will use AODV and sparse zones will use DSDV. Illustrated in Figure 9, RoVy works similarly to a "nested multi-hop" topology. Interoperable nodes are relays maintained by RoVy, running both AODV and DSDV situated at the edge of dense zones. Their role is to convert AODV data packets to DSDV and vice versa, similarly to a how an Ethernet GW work. A simulation was conducted with 1,000 nodes and a data payload of 140 bytes, which is within the maximum payload size of a SF8/SF7 LoRa network. Results show that in comparison to pure AODV and DSDV, route discovery using RoVy managed to maintain an average latency between 20 and 23 s, gradually increasing as the distance between nodes increases. Comparing pure AODV/DSDV with RoVy, RoVy saw general improvements with a 95% confidence interval. The authors have also reported an average end-to-end data latency of 110 ms but no direct comparisons were made.

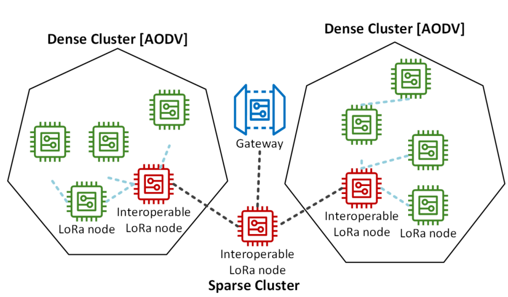

Fig. 9. Interoperation between active and reactive routing protocols; RoVy [17].

6.2 Reactive Routing Protocols

Huang et al. [38] modified the AODV protocol and simulated it on a LSN arrangement. RSSI is used as the routing metric instead of hop counts. A simulated scenario of 34 locations with a total length of 13.6 km resulted in a PDR from the GW to the furthest EN of up to 99%. Dwijaksara et al. [27] proposed another multi-hop protocol for inter-GW communications. Aslam et al. [13] also did a study on GW multi-hop, but that paper focused on power consumption. Dwijaksara et al.'s work categorised inter-GW communications as Class G, since LoRaWAN protocol specifications does not specify any G2G transmissions. The proposed protocol includes two functionalities of routing and channel access, where the routing protocol finds a route from an EN to the server (i.e., a list of forwarding GWs) and the channel access scheme specifies the packet transmission mechanism between two GWs. The protocol also has a packet aggregation feature that enables efficient G2G packet forwarding. Two categories of GWs were introduced; direct GW is directly connected to the NS through a backhaul link, and the intermediate GWs can only use the G2G communication for forwarding data to and from the NS. Each GW has 2 LoRa radios, one for communicating with the ENs and another to communicate with the other GWs. AODV is modified and used to utilise ToA as the route distance, since the goal was to determine the minimum ToA from intermediate GWs to direct GWs. LBT and packet aggregation were also utilised to reduce on-air packet collisions and ToA. Simulations were done comparing the baseline of G2G routing scheme and pure ALOHA-based channel access without packet aggregation. Results show that with G2G the packets spend less ToA when compared to single hop. The LBT-based channel access of the proposed scheme also incurs much fewer collisions as compared with pure-ALOHA, thus achieving high data throughput.

Yang et al. [88] addressed the challenges of LPWAN's highly non-isotropic coverage and the unpredictability of challenging environments by proposing a data payload ferry mechanism. Named FerryLink, it combines both RSSI sampling and **Channel Activity Detection (CAD)** for optimal data transmission. A transmitting EN first identifies itself that the GW connectivity is not reliable through the analysis of RSSI threshold values and using the CAD feature built into Semtech's LoRa chipsets to scan for surrounding ENs. Once a suitable candidate EN is found, the transmitting EN will send its data to the candidate receiving EN using downlink signals, thus leaving the uplink bandwidth and airspace channels clear. To reduce energy consumption, FerryLink assumes that both ENs are in good synchronisation after successfully getting in touch with each other for the first time. A total of 50 ENs were deployed, some in static environments, such as city buildings, and dynamic ones, such as on bicycles. Data were collected for one month. Results show that PDR for some ENs increased from under 10% in traditional LoRaWAN mode to nearly 100% with FerryLink. FerryLink only consumes about 30% less power when compared to LoRaWAN, due to lower SF multi-hop and increased PDR. No analysis was done for transmission delay.

Hong et al. [36] took the ideas of CT [28, 51, 89] and AODV and proposed a hybrid protocol called HBEE. The authors claimed that **Concurrent Transmission (CT)** works for LoRa despite a contradiction by Liao et al. regarding the increasing correlation between number of nodes and rate of unrecoverable packet collisions in a CT environment [51]. A lab experiment was conducted and identified that a PRR of 98.5% is still achievable due to the large link budget over the radio's receiving sensitivity. However, the assumption is that all ENs have good RSSI. HBEE implements zero-control-packet routing, learning routes by analysing the header of data packets without the presence of extra routing packets. HBEE uses two LoRa channels, public and private. An exploring message is flooded to the network via public channel using CT. This process segregates the network into hierarchical radii and enables the ENs to identify the suitable next-hop EN closer to the GW. While in transmission mode, the EN picks a channel, checks if the airspace is available, and then submits a handshake request containing the private channel details to the receiving EN. Receiving EN parses the handshake request, switches over to the private channel, and receives the data from the transmitting EN. This process repeats until data reaches the GW. Experimental comparison was done between synchronous (CT) and asynchronous (HWMP) in terms of initial flooding routing setup time, and results show that synchronous had theoretically no delay due to all nodes simultaneously flooding the airspace. HBEE shows better efficiency than AODV due to the inherent random possibility of AODV at creating imbalanced routes. PRR of 95.7% and 89.8% were achieved during network formation and normal operation phases, respectively. However, there are no PRR comparisons with other protocols.

Osorio et al. [66] evaluated a gossip-based mesh routing algorithm, MACGSP6 on LoRa with the goal of improving EN communications that have NLoS to the GW. MACGSP6 saves energy in delay-tolerant WSN applications through the improvement of flooding protocol resulting in the reduction of payload duplicates in the network. All ENs are in a reactive state; they only react once packets are received. MACGSP6 does not explicitly compute routes and instead, ENs opportunistically send the message to all other ENs that so happen to have their radios on at that point in time. An EN first transmits a packet, stays on, and when it hears the same packet from another EN the said transmitting EN will shut itself down, or else it will wait for a period of time and retransmit. The overhearing process repeats with each EN until data reaches the GW. RSSI and PDR were compared between traditional LoRa star topology and MACGSP6 through simulation with 1,000 ENs and on a university campus with 8 ENs. Data packets were able to reach the GW with the least number of hops. PDR for NLoS nodes increased from 0% in a star LoRa network to a minimum of 53% with MACGSP6. However, delay analysis shows that these type of protocols are not recommended to be implemented in time-sensitive systems as it induces propagation delay.

Feng et al. [30] addressed the limitations of GW coverage and power node exhaustion for single-hop networks by proposing a multi-hop protocol, MLoRa. The objectives are to provide a high node lifetime and low ToA solution. A novel metric was proposed: gain-to-cost ratio. It is defined by the ratio of an average amount of reduced airtime for transmission of a single packet over the sum of redundant transmissions due to relaying. A high ratio leads to a longer node's lifetime due to reduced ToA requirements. MLoRa operates on the WSMST algorithm and also implements a backtracking algorithm to recursively check the data payload of an EN with all other ENs on the upstream path to the GW. This is to ensure that the aggregated payload does not exceed the frame payload size and causing a link overload. A simulation shows that the nodes using MLoRa saw an average 10–30% increase in EN lifetime over traditional LoRa star topology, even up to 50% for a dense topology. It also shows that sparse topologies have up to four times higher gain-to-cost ratio over denser topologies as spare networks benefit more on multi-hop.

Muthanna et al. [62] presented MQ-LoRa, a multi-hop network with transmission policy enforcement and QoS. The objective of their research is to enhance the performance of LoRa IoT networks by incorporating deep RL algorithms for transmission policy enforcement and multi-hop routing. The aim is to optimise QoS metrics, such as packet delivery ratio, end-to-end delay, and energy efficiency to meet the requirements of various IoT applications. The entire network is segregated into smaller clusters and within each cluster, a Routing **Header Node (HN)** is elected. The purpose of a Routing HN is to collect data from other ENs and select the optimal multi-hop route to the GW. Two lightweight algorithms were proposed for optimising multiple route selections: Concurrent Optimisation Multi-hop Routing, which is based on the Mayfly insect pattern algorithm, and the Shuffled Shepherd Optimisation Algorithm. Both algorithms evaluate all route paths in parallel and rank them into three levels of priority based on their respective fitness value. Fitness value consists of latency, link quality, throughput, reliability, and energy consumption. A simulation comparison was then done to evaluate the performance among RT-LoRa, Multi-Hop, and, LoRa+. It was observed that the PRR of multi-hop, RT-LoRa, and LoRa+ were 40%, 30%, and 40% lesser than MQ-LoRa, respectively. Energy efficiency also increased up to 28% when compared to the three above-mentioned protocols.

6.3 Hybrid Routing Protocols

Lundell et al. [53] proposed a G2G mesh routing protocol to extend coverage. GWs without direct Internet access will forward packets to another GW that has proper Internet connectivity. They took the HWMP and AODV as blueprints, adapted them to LoRa, and built a packet tunnelling mechanism. This makes the adaptation transparent to both ENs and NS, ensuring compatibility with existing LoRaWAN standards. Laboratory experiments and field tests were conducted, with uplink messages managing to traverse a four-hop network. Downlink transmission was not tested. Other features present in HWMP and AODV such as route failure handling, route timers, and acknowledgements were not implemented as the authors goal was only to provide the initial proof of concept that G2G routing for multi-hop LoRa is feasible.

Kawabata et al. [42] proposed Convergecast, consisting of both asynchronous and synchronous protocols using duty cycling. The authors then took the ADCC scheme as described by Chen et al. [19] and compare its performance with Convergecast. In ADCC, every EN adaptively adjusts its duty cycle by comparing its own energy consumption with the largest energy consumption of the entire packet delivery flow. These data are stored in a feedback ACK packet generated by the sink node. Kawabata then introduced hybrid Convergecast to facilitate packet delivery. ENs that are close to the sink apply a synchronous protocol, and far nodes apply an asynchronous protocol. Routing of packets along hops is determined by residual voltage and energy consumption. The transmission delay for Convergecast is 1 s higher than ADCC but network lifetime increased by up to 6 times. Results also show that a synchronised protocol uses less energy but requires accurate time synchronisation and far ENs have a higher chance of packet transmission errors.

A high SF enables ENs to cover long distances, but ToA also increases, causing channel airspace congestion. Ahmar et al.'s work [3] addresses the issue by proposing a workaround. Since different SFs equate to different communication ranges, the authors proposed Smart-Hop. Its purpose is to selectively replace single-hop high SF links with chain links of lower SFs while having low overhead routing to minimise data extraction times. A duty cycle of 1% is adhered to as per regional regulatory requirements and time synchronisation is done similarly to Ebi et al.'s [28] works. Data aggregation was also applied to the PHY payload. To determine the maximum number of packets that can be aggregated, an algorithm considers both the payload size and the number of ENs of each route to ensure that this limit is not breached; e.g., at SF12, the maximum payload is 51 bytes and for SF7 it is 222 bytes. A test was done with 12 ENs and a single GW. Results show that data extraction time is reduced by around 64% in the best-case scenario and 8% in the worst-case when compared to a single-hop network. The average SF at which the network settled down to was SF7.

7 SECURITY

As specified in the official LoRaWAN protocol documentation, NS to EN communication channels are encrypted using AES 128. However, even with encryption LoRa-based networks utilising the LoRaWAN protocol are still subject to exploits such as flooding, replays, DoS, spoofing, and MITM [84]. In order for LoRa-based networks to be widely adopted, security needs to be one of the main priorities as emphasised by multiple authors in various papers. Although there are not many works done toward security on LoRa multi-hop networks, single-hop LoRaWAN protocol research establishes the foundation for understanding and addressing security vulnerabilities and threats. By examining the security mechanisms, protocols, and cryptographic algorithms in

Table 8. Previous Works Related with LoRa's Security

Ref.	Author	Year	Classification	Objective	Technology	Addressed Challenge	Outcome	Limitation
[12]	Aras et al.	2017	DoS	Exploit existing LoRaWAN specifications to facilitate jamming.	LoRaWAN	Availability and reliability	Jamming was successful around 40% of total attempts.	Evaluation only conducted in perfect scenario situations; long ToA and high RSSI.
[44]	Kim and Song	2018	Privacy	To enable secure inter-EN connectivity using industry standard encryption.	LoRa	Authentication and validation	ENs able to share cryptographic keys and encrypt/decrypt connections.	Battery consumption increase around 5%. Protocol overhead increased receive latency.
[56]	Mahdi et al.	2019	Privacy	Study on security vulnerabilities and their countermeasures in relation to smart city deployment.	MANET agnostic	Transmission reliability and resiliency	Modified Multipath Optimised Link State Routing (MP-OLSR) protocol to enhance security via multi-path and avoid packet loss using Expected Transmission count (ETX) metric instead of hop count.	Current simulation of MP-OLSR does not support load balancing but proposed as future works.
[82]	Tsai et al.	2019	Privacy	To reduce power consumption during encryption operations on ENs and propose new mutual authentication process.	LoRa	Authentication and encryption	External hardware encryption module reduced active and idle power consumption by 62 and 88.5% respectively, computational latency reduced by 38.8%. Proposed mutual authentication process resisted eavesdropping and replay attacks.	Power toggling of module introduced latency. Current serial computing architecture may cause reduced performance. Future proposal suggested parallel processing.
[35]	Hessel et al.	2020	DoS	Develop a test bench framework, ChirpOTLE and asses security of real-world networks.	LoRaWAN	Authentication and validation	Potential vulnerability in ADR mechanism identified and successfully exploited it to conduct a DoS attack	Scope is limited to the wireless link between ENs and GW. Full mitigation requires modifications to current LoRaWAN specifications.
[60]	Moraes and Conceicao	2021	Privacy	To improve data security in LoRaWAN-based applications that utilise non-trusted NS	LoRaWAN	Authentication and validation	Zero Trust policy proposed. The proposed payload format provided additional data confidentiality and integrity layers to mitigate risks. Applicable to LoRaWAN v1.0 onwards.	Overhead processing delay (based on 16-MHz Arduino) due to additional encryption range from 2.7 s for 1 byte of raw data to 31.2 s for 96 bytes.

single-hop scenarios, potential weaknesses can be identified and mitigated. This knowledge is crucial when extending the network to multi-hop scenarios, as the security measures implemented in single-hop LoRa networks may need to be adapted or enhanced to protect the network's integrity, confidentiality, and availability in a multi-hop environment. This also aids in understanding the new attack surfaces and potential threats that can exploit the additional complexity introduced by multi-hop communication. Table 8 is a summarised list of previous work on addressing LoRa's security challenges.

7.1 Privacy

Multi-hop LoRa require ENs to communicate with each other. However, current LoRaWAN protocol specifications only officially support communications between ENs and GW. Kim and Song [44] clarified that inter-EN communications are exposed to security threats as they are not encrypted and does not support basic security requirements such as mutual authentication, confidentiality, and validation. The authors proposed a secure link establishment scheme to protect this new dimension of communication thus allowing ENs to securely share cryptographic keys for encrypting connections. An experiment was done on a testbench to evaluate the time it took for an EN to transmit and receive as well as how much power it consumed; both with and without inter-EN encryption. The time to transmit variance is negligible for both with and without encryption. However, receive time increased by 35–40% due to additional overhead from NS to EN. Power usage increased by approximately 5% when compared with the basic protocol without security. However, the authors claim that the security benefit outweighs the negative impact on battery life and latency.

For a total end-to-end encryption for multi-hop networks, encryption beyond the nodes also needs to be considered. To reduce infrastructure costs, a third-party NS is usually used to facilitate the data transfer from GW to application servers. Moraes and Conceicao [60] raised the concern that the NS is usually not considered a trusted entity. Their work assessed the data security risk of using an untrusted NS in a LoRaWAN-based application by using **the National Institute of Standards and Technology (NIST)** guidelines: assessment preparation, threat evaluation, vulnerabilities identification, and risk level evaluation. The impact of using an untrusted NS may lead to corruption and disclosure of data and thus mitigation is required. They proposed a process of externally encrypting the frame payload without depending on the NS, by using a mechanism that is compatible with both LoRaWAN 1.0.x and 1.1. Figure 10 describes the frame encryption process. First, the data are hashed with SHA-256 as suggested by NIST. Then, the last four bytes of the hashed data are concatenated with the actual data and it is used as the 128-bit key and encrypts the data with AES-CTR and an **Initialisation Vector (IV)**. Finally, the last four bytes of the IV are concatenated with the encrypted bytes generated from the second step. The NS will never know the contents of the pre-encrypted payload. A test case experiment was conducted using LoRaWAN 1.0.x protocol on a 16-MHz Arduino microcontroller. Decryption and data validation was successful, leading to the conclusion that the vulnerability is no longer available. However, the encryption and validation overhead was an additional 2.7 s for 1 byte of raw data and up to 31.2 s for 96 bytes.

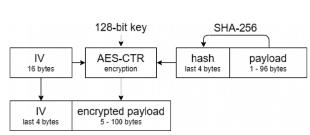

Fig. 10. Suggested encrypted payload to mitigate vulnerabilities due to using an untrusted NS.

Mahdi et al. [56] deduced that in order for smart cities to thrive, information security must be considered, as a reliable IoT communication infrastructure is a core concept of a smart city. Multiple attack vectors on MANET protocols have been identified, such as black/grey hole, Byzantine, wormhole, route poisoning, packet replay, and information disclosure. However, most of the existing proposals and solutions to address these issues require a decently reliable routing protocol. Hence, the introduction of MP-OLSR protocol, an RFC8218 standard that in turn is an extension of the original OLSR, RFC3626. MP-OLSR periodically sends out messages for routing table maintenance (nature of active routing) but not for all routes (nature of reactive routing). ETX is used as the metric instead of hop count. It is based on calculating the PDR so that routes will primarily choose the most reliable path. A simulation was done on NS2 to compare OLSR and MP-OLSR, both either using traditional hop count as the metric or the proposed ETX metric. Results show that in terms of end-to-end delay, original MP-OLSR is the lowest, whereas MP-OLSR with ETX is the highest. This is due to additional ETX calculations and path selection but as the delay variance between the slowest and fastest is less than 0.1 s, it can be considered as negligible. For PDR, MP-OLSR with ETX gave a consistent PDR of above 90%, especially with high density networks. However, the authors did not evaluate the modified protocol with the described network layer attack scenarios.

Tsai et al. [82] studied the **Advanced Encryption Standard (AES)** algorithm. It is used to secure end-to-end communications between ENs and NS, enabling mutual authentication, confidentiality, and integrity validation. LoRaWAN combines the original AES encryption and decryption algorithm with several additional operational modes; including a Cipher-based Message Authentication for data integrity and Counter mode for data encryption. The authors stated that AES introduces problems such as high amounts of processing and increased power consumption due to multiple cycles of repetition to complete the encryption/decryption process (as shown in Figure 11). Hence, they proposed the concept of **Low-Power AES Data Encryption Architecture (LPADA)**. It is a low-power hardware module consisting of Content Addressable Memory for the SBox (a lookup table for substitution values used for AES non-linear and invertible transformations) with power gating to toggle between active and standby modes. Simulations were conducted over several supply voltages and show that LPADA reduced active power consumption by around 62%, a reduction of up to 88.5% when in idle mode, and 38.8% reduction in terms of computational latency when compared to pure software AES processing. The reduction in energy consumption may be beneficial to multi-hop LoRa network topologies as the intermediate nodes will not only have the role on being an EN but also a relay for other nodes. This in turn relates to more frequent encrypting and decrypting of packets. With LPADA, the operational lifetime of battery operated intermediate nodes may be increased.

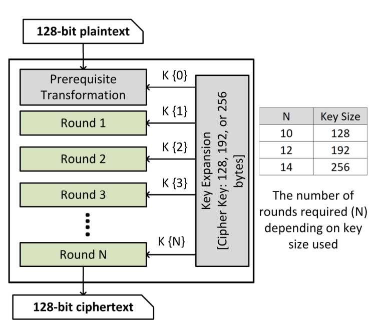

Fig. 11. The repeated nature of AES encryption.

7.2 Denial of Service

Despite LoRa having good noise rejection, its signals can still be jammed through the exploitation of coexistence with other CSS signals on the same channel. Known as DoS, intermediate paths between ENs and GW may be interrupted causing network disruptions. Attackers can deploy DoS attacks to destabilise the network or they may even execute advanced techniques such as deploying a nearby rogue node and then initiate DoS attacks to reduce the radio sensitivity of legitimate ENs, forcing them to connect to the rogue node. Once a rogue node has infiltrated the network, secondary attacks are now possible. Hessel et al. [35] compared CSS with ultra-narrow band modulation and showed that LoRa is susceptible to inter-interference. They proved that interference between LoRa frames of different SFs is low while signals of the same SF and channel has to be at least 6 dB stronger to cause any negative effect. Aras et al. [12] used this knowledge to create a LoRa jammer. This device is able to perform triggered, payload-based reactive and targeted jamming. Successful tests were conducted at two different locations. The authors also discovered that LoRaWAN frames do not have any temporal information, lowering the complexity of replay attacks. Using that knowledge, they demonstrated a wormhole with two nodes. As shown in Figure 12, a wormhole attack consists of a minimum of two malicious nodes at both ends, connected via a faster out-of-band line. Packets received or sniffed from one end are then tunnelled over to the other malicious node at another point, manipulated, and then broadcasted to the unsuspecting ENs, causing a break in data consistency. Hessel et al. [35] also introduced a security evaluation framework to address risk assessment and interoperability complications, called ChirpOTLE; testing attacks in testbed scenarios and assessing the outcomes in real-world environments. The authors have demonstrated their framework and implemented a DoS attack targeting the vulnerabilities of LoRaWAN's ADR feature. The attack featured a wormhole between an EN and a GW, and spoofed the ADR protocol using manipulated SNR and RSSI values via replay attacks. This resulted in messages on both sides to have lost their relations causing legitimate packets to be dropped. An update in LoRaWAN's specifications was proposed to address the issue and the immediate workaround was to deploy more GWs but that will be in contradiction to the goal of a distributed LoRa multi-hop topology. Beacon spoofing was also demonstrated on Class B devices (a similar discovery made by van Es et al. [84]), and ENs were lured to follow the spoofed beacons, resulting in a DoS attack. A specification update was also proposed to address this issue as the suggested workaround does not mitigate beacon spoofing but only made it more difficult to do so.

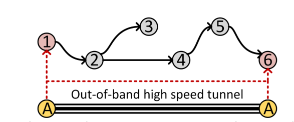

Fig. 12. Wormhole attack using a faster out of band communication.

8 OPEN ISSUES

8.1 Correlation between Power Consumption and LoRa Parameters

Power consumption is one of the top challenges faced in maintaining extended lifetime for all remote sensors. Referring to the analysed literatures, a number of notable researchers have conducted experiments and proposed solutions to optimise power consumption for LoRa networks, regardless if they are the traditional star topology or multi-hop and mesh networks. Works from Anedda et al. [11], Aslam et al. [13], Grochla et al. [31], Gupta and Snidgh [32], Lumet et al. [52], and Paul [70] led to a consensus that despite having more nodes, a multi-hop LoRa network does lead to lower average power consumption when compared to a typical star network. LoRa has two parameters that influence the overall throughput and distance a data signal can travel: BW and SF [75]. These two correlate with ToA of a packet, and thus power consumption. Therefore, the potential of mass deploying sensor nodes at vast distances is achievable with a minor compromise toward energy wastage. Parameter combination and optimisation as studied by Dix-Matthews et al. [26] also contributed to better power efficiency through the utilisation of "best-fit" parameters in obtaining the best performance out of the LoRa radios. However, increased architectural complexities uncover multiple challenges such as equipment upkeep, increased point of failures and strategies of intermediate node placements. Furthermore, node health monitoring and dynamic parameter selection for optimised performance lead to the challenge of management and coordination mechanisms.

There is a need for comprehensive empirical studies and modelling to better understand the intricate relationship between power consumption and LoRa parameters. Conducting extensive experiments in real-world deployments across various environmental conditions, topologies, and traffic patterns will help quantify the impact of different parameters on power consumption. Such studies can provide insights into the tradeoff between transmission range, data rate, and power consumption, aiding in the development of energy-efficient LoRa parameter configurations. Second, research efforts should focus on developing advanced power management techniques and algorithms tailored specifically for multi-hop LoRa networks. This includes investigating adaptive power control mechanisms that dynamically adjust transmission parameters based on factors such as link quality, node distance to the gateway, and available energy. Designing intelligent algorithms that balance the tradeoff between power consumption and network performance metrics, such as latency, reliability, and throughput, is an interesting research direction. Additionally, exploring energy harvesting and energy-efficient routing strategies can contribute to further optimising power consumption in multi-hop LoRa networks.

8.2 Capacity Planning

Capacity planning is an essential process for any industry that has the intention to adopt and deploy a certain technology. One of the key challenges is optimising network coverage and capacity while minimising interference. Research should focus on developing efficient placement algorithms for nodes to ensure adequate coverage and signal strength throughout the network. Additionally, investigating interference mitigation techniques such as adaptive power control, frequency hopping, and advanced channel assignment strategies can help maximise network capacity by minimising interference and collisions. Addressing scalability is another critical aspect of capacity planning in multi-hop LoRa networks. As the number of nodes and the complexity of the network increase, it becomes challenging to manage and effectively route traffic. Research should focus on developing scalable routing algorithms and protocols that can handle the increased load and provide efficient data delivery. This includes exploring adaptive routing strategies, load balancing mechanisms, and hierarchical approaches to distribute the traffic intelligently and optimise the network's overall capacity. Furthermore, incorporating machine learning and artificial intelligence techniques into capacity planning can be a promising research direction. By leveraging historical data, network statistics, and predictive models, intelligent algorithms can assist in predicting network traffic patterns, optimising resource allocation, and making proactive capacity planning decisions to ensure efficient network operation and improved performance.

8.3 The Choice between Payload Aggregation and Compression

Other than SF and BW, payload size can also be manipulated. Table 9 describes the relation between SF, BW, bit rate, and frame payload size. By being able to fully utilise a single payload packet, power efficiency can be improved as the ratio between data over the number of packets required increases. As LoRa's potential target market is WSNs, the probability of sensor readings following a linear pattern is high and this is what Vaananen and Hamalainen's work [83] took advantage of. A LoRa radio that consumes a definite amount of energy for a specific parameter configuration and when the parameters are already in an optimised state, compression algorithms may potentially contribute to further improving energy efficiency. Most sensors record readings that tend to behave linearly (e.g., humidity and temperature). By recommending the implementation of linear-based compression algorithms, a high compression ratio can be achieved leading to

Table 9. List of LoRa Payload Size in Relation to Modulation and Bandwidth

Data Rate	Configuration		PhysicalBit Rate	Maximum Payload Size (MAC)	Maximum Payload Size (Frame)
	Modulation	Bandwidth			
0	SF12	125 kHz	250	59	51
1	SF11	125 kHz	440	59	51
2	SF10	125 kHz	980	59	51
3	SF9	125 kHz	1,760	123	115
4	SF8	125 kHz	3,125	230	222
5	SF7	125 kHz	5,470	230	222
6	SF6	250 kHz	11,000	230	222
7	FSK, GFSK	150 kHz	50,000	230	222
8–15	Reserved for Future Use				

lower data transmission periods and ToA of a transmitted data packet. Data aggregation as studied by Paul [70], and Gupta and Snigdh [33] has shown improvements in EN lifetime and lowering transmission time. Pairing aggregation with compression, higher power efficiency is achievable, since more data can be squeezed into a single payload. Exploring the efficiency of payload compression techniques for multi-hop LoRa networks is another important research direction. Developing lightweight compression algorithms that can achieve a good compression ratio without imposing significant processing overhead is crucial. Additionally, evaluating the compatibility of compression techniques with the specific data types and characteristics in LoRa networks is necessary. Research should aim to develop optimised compression algorithms that can effectively reduce the payload size while considering the limited resources and low-power nature of LoRa nodes.

To achieve even higher power efficiency, time scheduling is also feasible but with certain disadvantages such as latency [1] where data packets will first have to be queued up and only transmitted once either a timeout or queue length has been reached. This is a disadvantage for latency-sensitive applications. Research should focus on finding the optimal payload size that minimises overhead while balancing the impact on latency. This involves investigating adaptive aggregation algorithms that can dynamically adjust the payload size based on network conditions, traffic patterns, and application requirements. Investigating the combined use of aggregation and compression techniques is also an interesting research direction. Assessing the benefits and tradeoff of using both approaches simultaneously can lead to further optimisation of resource utilisation and network capacity. Developing hybrid schemes that are able to adapt between payload aggregation and compression based on network conditions, traffic load, and energy constraints can contribute to more efficient and flexible data transmission in multi-hop LoRa networks.

8.4 Synchronised and Asynchronised Transmissions

An EN is usually powered from a constrained source such as a battery and its lifetime is highly dependent on the power source's capacity. Therefore, the energy savings of an EN is very important. As duty cycling is primarily for regional regulatory restrictions, it has also attracted much attention as an evolution for improving energy savings. One of the main factors that wastes energy is idle listening, where node radios are actively powered on but without any radio activity. Airspace density is also a challenge. A study conducted by Lavric [48] shows that the maximum number of LoRa sensors that can communicate on the same channel and SF is 1500. However, to maximise performance, proper planning and scheduling are required. Without them, PDR will be drastically affected, rendering the network inoperable. A popular question that researchers encounter regarding multi-hop LoRa is “How to ensure that the transmitter is able to find the receiver at the

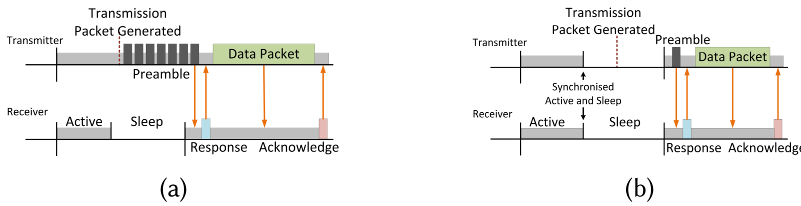

Fig. 13. Two different types of node synchronisation protocols; (a) asynchronous and (b) synchronous.

right time and channel while ensuring long network lifetime and compliance with government regulations?"

Kawabata's work [42] shows the possibility of asynchronised and synchronised transmissions (Figures 13(a) and (b)). Synchronised transmissions are the most common among the works being researched. Nevertheless, it requires strict time syncing among all ENs and the probability of sync errors for ENs that are far away will be higher due to uncontrollable factors such as interference, weather, and dynamic obstacles. Ebi et al. [28] proposed a time synchronising method using timestamps. The initial UTC time was from an external source such as GPS or DCF77 radio and is embedded into the frame payload. Whenever time synchronisation is required, the network will be flooded with sync data and the actual data will have to be paused, a limitation that requires further addressing. Bezunartea et al. [15] used a Semtech SX1272 module for reference and identified that due to crystal clock drift, a beacon is needed to re-synchronise its clocks for about every 1,000 slots when a guard time of 22 ms is used. With a 1% duty cycle regulation, this equates to only 1 in 100 allowable slots, a significant 10% bandwidth overhead. As shown by the above-mentioned works, achieving synchronisation among nodes in a multi-hop LoRa network is challenging. Synchronised transmissions can improve interference mitigation and overall network performance by avoiding collisions and optimising spectrum utilisation. However, synchronising nodes in a distributed and dynamic network is complex. Research should focus on developing efficient synchronisation protocols and algorithms that can handle the timing constraints of multi-hop LoRa networks. This involves investigating synchronisation techniques that can adapt to network dynamics, node mobility, and varying propagation delays.

Exploring the advantages and disadvantages of synchronised and asynchronised transmissions is an important research direction. Synchronised transmissions can provide enhanced reliability and interference mitigation but introduce additional overhead and coordination complexity. However, asynchronous transmissions offer flexibility and allow for opportunistic data transmission but can be more prone to collisions and interference. Research should aim to evaluate the impact of synchronisation on network capacity, latency, energy consumption, and overall performance in multi-hop LoRa networks. Investigating adaptive strategies that dynamically select between synchronised and asynchronous transmissions is another promising research direction. Designing algorithms and mechanisms that can dynamically adapt the transmission mode based on network conditions, interference levels, and traffic patterns can lead to improved network efficiency and performance.

8.5 Dilemma of Concurrent Transmissions

One of the quickest ways to implement MAC scheduling mechanisms is by adapting known mechanisms from other LPWAN technologies and evaluate their feasibility on LoRa. Such examples are Offset-CT [51], Convergecast (from ADCC) [42], TDMA [28], and TSCH [15]. While CT may be beneficial for technologies such as Wi-Fi (known as MIMO) it may not be suitable for WSNs that include LPWANs due to higher power consumption. Figure 14 shows a total of

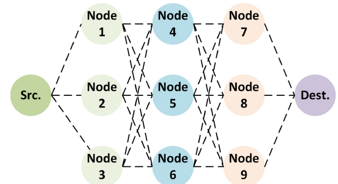

Fig. 14. Dilemma of concurrent transmissions in a multi-hop LoRa network.

11 nodes. With CT on LoRa, for every packet of data sent from source to destination, there will be a total of twelve transmission windows and twelve receiving windows, whereas if a routing protocol is used, then only four sets of transmission and receiving windows will be used. This severely limits the scalability of a LoRa network as the complexity exponentially increases with the number of ENs. A transmitting LoRa radio consumes a lot of power as well, shortening the lifetime of a network. Even if random transmission offsets have been implemented, packet collisions still exist if the node density is high causing PRR to drop significantly. Therefore, research should focus on designing energy-aware protocols and scheduling mechanisms that optimise energy consumption while considering the benefits and challenges of concurrent transmissions. Exploring energy-efficient routing strategies and transmission power control techniques can contribute to minimising energy consumption in multi-hop LoRa networks.

TDMA scheduling, however, has its own sets of challenges. During the time synchronisation and routing table flooding stage, the entire network will be saturated with data packets. This is inefficient, consumes more power, increases TX/RX overhead and in extreme cases, destabilisation on a network if the number of ENs is large due to airspace congestion. The number of ENs associated with a network will also be limited because of payload size. At SF7, the frame payload size is 222 bytes and at maximum SF12, it is only 51 bytes. Accurate timing synchronisation is essential to avoid collisions and optimise the utilisation of the shared medium. Researchers should explore synchronisation mechanisms that can handle the timing constraints of concurrent transmissions in multi-hop LoRa networks. Developing efficient synchronisation protocols and algorithms that can adapt to network dynamics, node mobility, payload restrictions, and varying propagation delays is a reasonable research direction to ensure successful concurrent transmissions.

8.6 Network Segregation and Far Nodes

One potential method that might improve the performance of a dense multi-hop LoRa network is to segregate it into smaller sub-networks, similarly to how a Virtual Local Area Network works. This reduces the chances of packet collisions and also shrinks the broadcast domain of a network. In LoRa network's context, segregation is possible through the usage of different SFs as shown by Zhu et al.'s paper [89]. However, the automatic segregation of nodes by SF also produces its own challenges. One such challenge is on determining the optimal clustering strategies and mechanisms. Research should focus on developing efficient algorithms for network segregation that consider factors such as node density, geographic distribution, and traffic patterns. These algorithms should aim to balance the load among clusters, optimise routing paths, and minimise interference to ensure reliable communication in multi-hop LoRa networks. Far nodes face challenges such as weaker signal strength, higher path loss, and increased susceptibility to interference. Researchers should investigate techniques for improving connectivity and data delivery to far nodes in multi-hop LoRa networks. This involves developing advanced routing algorithms that can handle longer hop distances and optimise transmission power settings. Additionally, exploring solutions like distributed intermediary nodes or cooperative communication techniques

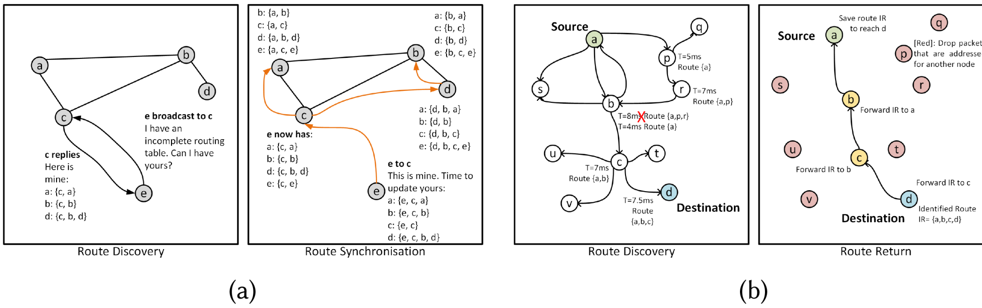

Fig. 15. Two different types of route construction processes on WSNs; (a) Proactive and (b) Reactive.

may assist in extending the coverage and reachability of far nodes. Designing energy-efficient strategies to support communication with far nodes is also a research direction to address the tradeoff between coverage and energy consumption.

8.7 Determination of a Suitable Routing Protocol for a Specific Deployment

Proactive protocols constantly share network topology between nodes. These protocols are suitable for large and dense networks with sporadic traffic but may also be suitable in sparse networks as demonstrated by de Farias Medeiros et al. [25] and Muthanna et al. [62]. While the data obtained from those studies show that the performance gain for a LoRa network is desired but in reality, they may not be ideal due to other factors such as EN lifetime. LoRa networks generally tend to contain a lesser number of dynamic ENs and mostly geographically stationary devices. Thus, the constant overhead of choosing suitable intermediate nodes and updating network topology may not be necessary for a LoRa's multi-hop deployment. As shown in Figure 15(a), for proactive protocols each EN will keep a copy of the entire routing table. When an EN wants to transmit, it refers to the routing table and selects the next hop node, reducing the need to broadcast request packets. This routing table is synchronised at specific intervals. If the nodes are highly mobile, then the routing table will be invalid causing affected ENs to incorrectly route due to broken routes. A workaround will be to reduce the routing table synchronisation interval, but that will cause airspace congestion and shorten the lifetime of all ENs due to extended radio operations. More computing resources such as CPU and RAM will also be required to maintain the routing tables as the network becomes denser. However, if the ENs are static and connected to reliable power sources, then active routing protocols are feasible.

Reactive protocols, however, are more suitable for WSNs. It only requires the sharing of topology when either a new route needs to be established or an existing route has failed. This leads to reduced control overhead, allowing more allocation for payload data and reduced power consumption. Figure 15(b) illustrates an overview of how a reactive protocol work. Most of the identified works used a reactive protocol, AODV being the most common [27, 38], since it has the tendency to maximise performance with low computing complexity [25]. However, reactive protocols do have their own challenges. During route discovery, the airspace may be congested with **Route Request (RREQ)** packets, affecting the reliability of receiving transmissions due to airspace collisions. Despite studies using CT as conducted by Liao et al. [51], Zhu et al. [89], Ebi et al. [28], and Hong et al. [36] presenting the benefits of airspace flooding, more research is needed as there are still insufficient scenarios that may influence the performance and reliability such as environmental conditions and urban topologies.

A hybrid protocol is generally categorised as a combination of active and reactive protocols. However, that term is not standardised and it can also mean a mix of any two or more routing protocols, regardless of active or reactive. Studies on utilising the benefits of multiple protocols consisting of either selectively switching between different protocols [17] or statically segregating them apart [53]. Carvajal-Gómez and Rivière [17] also highlighted the limitations of various protocols and apply an adaptive approach. While their proposed design does improve network performance, the disadvantages still exist. Having a multi-hop LoRa network with multiple protocols means protocol translations will be required, which equate to additional delays, since processing is required to decapsulate the packet and re-encapsulate to the destination protocol and vice versa. The router nodes are also required to have more computational power to perform these translation tasks. In extreme cases, the route node may lead to network destabilisation either due to battery exhaustion, node damage, or node theft. Routing protocols are beneficial but sometimes only a simple solution is required. FerryLink [88] and Smart-Hop [3] are some examples whereby multi-hop is still required but unnecessary for a full-featured routing protocol. Yang et al.'s [88] FerryLink utilises opportunistic routing approach on relaying data back to the GW. Ahmar et al.'s [3] work primarily maintains a LoRa star topology and only breaks down certain sections of the star topology into multi-hop, with the goal of keeping low SFs. Osorio et al.'s [66] MACGSP6 implementation on LoRa is simple to implement, does not require creating or updating routing tables, and has design similarities to ALOHA.

Research should focus on evaluating and comparing existing routing protocols, to identify their strengths and limitations in the context of multi-hop LoRa networks. Additionally, developing customised or hybrid routing protocols that address the unique characteristics of LoRa, such as long-range communication, limited bandwidth, and low-power operation, is an important research direction. Research should also explore adaptive and self-healing routing mechanisms that can dynamically adjust routing decisions based on network conditions. This includes investigating routing protocols that consider link quality, energy levels, and congestion in selecting the most suitable paths while maintaining fairness in the utilisation of ENs. Developing efficient route maintenance and repair mechanisms to handle node failures, mobility, and network dynamics is also a research direction to ensure reliable and efficient data delivery in multi-hop LoRa networks.

8.8 Radio Interference

Despite LoRa's CSS modulation having a high resistance to interference, LoRa is still subject to that phenomenon. There are three major types of interference that can degrade a LoRa network; co-channel interference, intra-interference, and inter-interference. One example of co-channel interference is the near-far effect. It is an effect where a data packet transmitted from an EN far away from the GW encounters airspace collisions from another data packet of an EN situated closer to the GW, or any surrounding interfering ENs that have at least 6 dB stronger RSSI than the far EN. This negative effect has been demonstrated by Aras et al. [12], van Es et al. [84], Mahdi et al. [56], and Hessel et al. [35] through the usage of wormhole attacks. Suggested workarounds include denser GWs, and higher SFs to improve sensitivity and increase interference resiliency. However, these suggestions are not beneficial for multi-hop networks, because one of the main goals is to extend coverage where existing telecommunication infrastructure is inadequate while maintaining a long network lifetime. ALOHA is also subject to both inter- and intra-interference. Other technologies that operate on the same ISM channel also interferes with LoRa signals, known as inter-interference, demonstrated using Choi et al.'s [21] airspace sniffer, LoRadar. Signal interference may be induced not due to external factors but also from within the electronic circuits themselves. This is known as intra-interference and can disrupt the efficiency and performance of LoRa networks. Research should focus on understanding the characteristics of different sources of interference, such as coexisting wireless networks, neighbouring LoRa networks, or other RF devices, and their effects on LoRa communication. This involves investigating interference detection and mitigation techniques, such as spectrum sensing, interference cancellation, or frequency hopping, to minimise the impact of interference on the network. Investigating adaptive transmission power control and channel assignment strategies can also help mitigate interference and optimise the network's overall performance. As nodes in the network communicate over multiple hops, interference can propagate and accumulate, leading to degraded signal quality and increased packet loss. Research should focus on developing interference-aware routing protocols and algorithms that consider interference levels, signal to interference and noise ratio, and link quality metrics during route selection and maintenance. Investigating interference-aware scheduling and MAC mechanisms can also help improve the efficiency and fairness of channel access in the presence of interference. Furthermore, exploring interference coordination techniques, such as interference-aware transmission scheduling or cognitive radio approaches, can enable efficient spectrum utilisation and interference management in multi-hop LoRa networks.

8.9 The Balance between Security and Performance

There is always a tradeoff between security and performance. To maximise throughput and battery life, restrictions in terms of CPU processing, RAM, and power consumption are required. However, cryptography mechanisms to facilitate encryption and decryption inherently consume more computing resources and energy. Kim and Song [44] identified that inter-EN communications require encryption and addressed it by proposing a link establishment scheme using AES encryption. As a result, power consumption increased by 5%. Tsai et al. [82] also deduced that AES encryption/decryption processes will consume a lot of computing resources and proposed that the AES processes be offloaded to a separate, dedicated hardware. They have managed to achieve up to 62% reduction in power consumption but at the cost of slightly increased latency due to additional programming calls to power toggle the hardware module. Therefore, the suggested research direction is to focus on developing lightweight and energy-efficient security mechanisms that provide strong encryption, authentication, and integrity protection while minimising the computational overhead. This includes exploring optimised cryptographic algorithms, key management schemes, and secure routing protocols that strike a balance between security requirements and the limited resources of LoRa devices. Additionally, investigating security-aware optimisation techniques, such as adaptive security configurations based on network conditions or traffic patterns, can help optimise the tradeoff between security and performance in multi-hop LoRa networks.

The understanding of vulnerabilities and threats specific to multi-hop LoRa networks is crucial in addressing the compromises between security and performance. Research should focus on identifying and analysing potential attack vectors, such as node compromise, data tampering, or replay attacks that can exploit the characteristics of multi-hop communications. This may involve developing intrusion detection and prevention mechanisms tailored for LoRa networks and investigating techniques for secure and reliable localisation of nodes in the network. Furthermore, exploring anomaly detection and behaviour profiling methods through machine learning can aid in detecting and mitigating security breaches while considering the performance implications. Developing comprehensive security guidelines and best practices for multi-hop LoRa networks can also help network administrators and designers strike an optimal balance between security and performance.

9 RECOMMENDED OPPORTUNITIES AND INDUSTRY APPLICATIONS

The practical use of multi-hop LoRa technology faces several key challenges. First, its limited bandwidth and data rates in unlicensed frequency bands can be a drawback for applications requiring high-bandwidth or real-time communications. Network scalability can be a hurdle as deploying a dense network with sufficient coverage requires significant investments in infrastructure and planning. Furthermore, LoRa's performance can be affected by environmental factors and interference, which can limit its range and reliability. Last, balancing power constraints with desired data rates and transmission ranges can be challenging for low-power, battery-operated devices.

However, when compared to other LPWAN technologies like Sigfox and NB-IoT in industrial applications such as smart farming and smart cities, LoRa offers several competitive advantages. First, deploying a LoRa network infrastructure can be more cost-effective due to the utilisation of unlicensed frequency bands, eliminating the need for expensive spectrum licenses. Second, when compared to cellular-based technologies like NB-IoT, LoRa also consumes less power. This enables extended battery life that is beneficial for remote or power-constrained environments. Third, LoRa offers flexibility in terms of data rate, range, and network configuration, allowing customisation to meet specific application requirements. In contrast, Sigfox provides a standardised network with limited customisation options. Last, LoRa surpasses other LPWAN technologies by having a more extensive ecosystem and receiving strong support from a thriving community. This results in a wealth of resources, development tools, and community knowledge available to developers and solution providers, ultimately facilitating seamless integration and streamlined solution development in industrial applications.

9.1 Smart Farming

Multi-hop LoRa plays a pivotal role in revolutionising the landscape of smart farming by enhancing connectivity and data collection across vast agricultural spaces. Farmers are able to embrace the concept of data-driven farming as they are able to access any type of information required to improve their business and sustainability, allowing farmers to manage their assets as efficiently as possible [46]. Multi-hop LoRa enables the extension of network coverage, which is particularly crucial in the context of expansive farms or plantations. Traditional single gateway LoRa networks have limited range, making it challenging to cover large areas comprehensively. Multi-hop LoRa addresses this limitation by allowing data to hop between nodes or sensors before reaching the central GW. This means that even remote sections of a farm can be included in the network, ensuring that no part of the agricultural operation is left unmonitored. This expanded coverage facilitates real-time monitoring of environmental conditions, crop health, and livestock well-being across the entire farm, enabling farmers to make timely and data-driven decisions [40].

Second, multi-hop LoRa contributes to the scalability and cost-effectiveness of smart farming solutions. With its ability to relay data over multiple hops, it reduces the need for additional GWs or infrastructure. This scalability is essential for farmers looking to expand their operations without incurring substantial infrastructure costs. By simply adding more nodes or sensors to the network, farmers can seamlessly scale up their data collection capabilities. This makes smart farming solutions more accessible to a broader range of agricultural businesses, from small family farms to large commercial enterprises.

Last, multi-hop LoRa enhances the resilience and reliability of data transmission in smart farming. Agricultural environments can be challenging, with obstacles such as terrain, buildings, and vegetation affecting wireless connectivity. Multi-hop LoRa networks are designed to handle such obstacles by allowing data to find alternative paths through the network, ensuring that critical information is reliably transmitted even in adverse conditions. This resilience is particularly vital for time-sensitive applications like early disease detection, where any data loss or delay could have significant consequences for crop or livestock health.

9.2 Smart Cities

The concept of smart cities may differ between people but most will agree on the general objectives; to improve the quality of life of the city residents through the reduction in waste, consumption, operating costs, optimising resources, and enhancing the overall operational performance of a city. The ideologies and concepts are to promote a modern and intelligent transformation toward urban planning, utilities, construction, and management through the usage of Information Technology, which can encompass big data, IoTs, geographical information systems and cloud computing [54]. One of the primary contributions of multi-hop LoRa to smart cities is its ability to create extensive and resilient low-power, wide-area networks. These networks can cover large urban areas and overcome the challenges posed by buildings, infrastructure, and other obstacles that often disrupt wireless communication in dense urban environments. By allowing data to hop between nodes and sensors, multi-hop LoRa ensures that even hard-to-reach or isolated areas within a city can be integrated into the smart city network, facilitating real-time data collection and monitoring.

In a smart city context, multi-hop LoRa supports a wide range of applications that can contribute to urban sustainability and efficiency [40]. For instance, it enables smart parking systems that help drivers find available parking spaces, reducing traffic congestion and emissions. Additionally, multi-hop LoRa can be used for waste management, allowing smart bins to signal when they need to be emptied, optimising collection routes, and reducing operational costs. Furthermore, it can support environmental monitoring by collecting data on air quality, noise pollution, and other factors, which is critical for informed urban planning and public health.

Moreover, multi-hop LoRa technology aids in the deployment of smart street lighting systems. By connecting streetlights to the network, cities can remotely monitor and control their lighting infrastructure, adjusting brightness based on real-time data such as pedestrian and vehicular traffic, weather conditions, and daylight levels. This not only enhances energy efficiency but also contributes to citizen safety and reduces light pollution. In summary, multi-hop LoRa is a foundational technology for building resilient and extensive IoT networks in smart cities, enabling a wide range of applications that enhance urban sustainability, efficiency, and the overall quality of life for residents.

9.3 Smart Industry

The gradual advancements and acceptance of IIoT technologies, including multi-hop LoRa networks, is indeed the foundation for Industry 4.0 and the evolution of smart industries. In today's industrial landscape, the demand for innovative solutions to enhance operational efficiency, optimise resource utilisation, increase productivity, and ensure environmental conservation and worker safety is paramount. Within industries like oil and gas, which traditionally relied on Supervisory Communications and Data Acquisition systems based on the ISA-95 model, there is a significant shift toward Industry 4.0 and IIoT [5]. This transition leverages new technologies to create a more interconnected and data-driven ecosystem. Multi-hop LoRa has emerged as one of the game changers in this context due to its ability to propagate RF signals effectively even within dense and challenging environments commonly found in processing plants, storage facilities, oil fields, platforms, and refineries.

One potential application that demonstrates the value of multi-hop LoRa networks in the oil and gas industry is pipeline monitoring. These pipelines can stretch across vast distances, sometimes hundreds or even thousands of kilometres, making continuous monitoring a complex task. Multi-hop LoRa networks utilising a linear topology can be strategically deployed along these pipelines, allowing data to hop from one node to another until it reaches a GW [1]. This approach extends the coverage range while ensuring data reliability and integrity. With multi-hop LoRa, GWs can be sparsely placed, reducing infrastructure costs, and simultaneously maintaining comprehensive coverage. Furthermore, in the context of oil and gas pipeline monitoring, multi-hop LoRa networks can monitor critical parameters such as gas/oil pressure, volume, flow rate, and pump health. Real-time data collection and analysis enable early detection of anomalies, leaks, or equipment failures, which can be critical for preventing accidents, minimising downtime, and ensuring the safety of personnel and environmental protection.

9.4 Smart Utilities and Metering

Utility companies providing electricity, water, gas, and heating services are currently facing a set of evolving challenges driven by the growing demand for efficient, high-quality, and reliable services in both urban and rural areas. To meet these demands while staying competitive, these utility companies are undergoing digital transformation efforts. One aspect of this transformation is the integration of monitoring and analytics into their operations, such as smart meters, substations, and transmission lines. These technologies play a pivotal role in reducing response times to issues and minimising service outages, ultimately improving the overall reliability and quality of utility services [38].

In the case of water utility companies, the adoption of smart meters equipped with multi-hop LoRa technology may provide several advantages. Remote meter readings enhance billing accuracy, providing customers with more precise usage data, and allowing utility companies to identify and address issues more efficiently. Furthermore, these meters can provide real-time alarms and notifications, alerting utility companies to potential problems or irregularities, which can help prevent accidents and improve safety. Similarly, water utility companies can benefit from smart meters in various ways. Besides improving billing accuracy and usage monitoring, smart meters can potentially be used for leak detection [47]. By continuously monitoring water flow and pressure, these meters can quickly identify leaks in the distribution network, allowing for timely repairs and substantial water conservation.

Additionally, multi-hop LoRa technology, particularly in mesh topologies, plays a significant role in facilitating the utility companies' transition to Industry 4.0. This is especially crucial in rural areas where network infrastructure coverage can be a challenge. The mesh-enabled LoRa networks can extend the range and coverage of data transmission, ensuring that even remote and underserved areas can be integrated into the utility's monitoring and control systems. The feasibility of multi-hop LoRa has been demonstrated in various real-world applications, such as rural microgrid projects, as exemplified by the study conducted in Nepal by Marahatta et al. [57].

10 CONCLUSION

Demand for the deployment of efficient and cost-effective LPWANs for IoTs is becoming more apparent due to the constant growth of globally connected devices and their ever-growing variation of application requirements. The existing survey papers on addressing the challenges of deploying multi-hop and mesh LoRa networks are still lacking, which led to the review of recent papers published between 2015 and 2023, and proposing a classification of four (4) aspects that determine the feasibility, functionality, and practicality of multi-hop and mesh LoRa networks: (1) Energy awareness, includes parameter selection, distance, and clustering; (2) concurrent access and duty cycle regulations encompasses network segregation, and time synchronisation; (3) routing protocols covers proactive, reactive, and hybrid routing protocols; and (4) security, consisting on privacy, and DoS. An architectural overview of LoRa and LoRaWAN is presented, including classifications, state-of-the-art methods, and solution methodologies of the abovementioned aspects. A detailed discussion of the aspects' issues and recommended industry applications that will benefit from multi-hop and mesh LoRa networks are also provided. Findings show a consensus that multi-hop networks and adherence to radio duty cycle are able to lower average power consumption. Concurrent access method via network flooding can improve network performance but more research is needed due to different conclusions. Reactive protocols have a slight advantage over proactive ones in terms of node deployment scalability and lower power consumption due to less frequent routing table updates. Hybrid protocols are only beneficial in specific topology deployments. Security is still a major concern for LoRa networks, particularly on attacks such as MITM, spoofing, wormholes, flooding, packet replay, and information disclosure. The upcoming trend of implementing machine learning algorithms to multi-hop and mesh LoRa networks opens up a wide range of possibilities, ranging from airspace efficiency, efficient route selection, and improving data throughput. The aim of the proposed classifications is to encourage researchers to discover, architect, and implement new and efficient solutions by providing suggested directions to assist in improving the efficiency of LoRa-based multi-hop and mesh networks.

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